Plot Delineation

APPN Aerial SOP — Locked revision 1.03

APPN – Plot Delineation

This protocol defines the APPN standard for plot delineation shapefiles — their structure, attributes, and storage location within the APPN folder hierarchy — and documents the supported methods for producing them from APPN aerial imagery (typically RGB orthomosaics captured by GRYFN UAV systems). Consistent plot delineation underpins repeatable phenotypic analysis across APPN trials.

Important

The APPN plot shapefile specification below must be followed for all trials, regardless of which method is used to generate the shapefile. For any deviations from this specification or these methods (e.g. alternative tools, non-standard plot layouts), keep detailed records of the changes made, the rationale, and any implications for downstream analysis.

Important

Document status — methods under cross-validation. The APPN plot delineation standard and the three supported methods below reflect the decisions made at the **APPN Field EWG plot-delineation meeting**. **Zeljana Grbovic** is currently testing whether all three methods (FIELDimageR, DPIRD Field Mapping Tool, and GPT plot creation tool) produce consistent results. Parts of this protocol may change depending on the outcome of that testing, which is tracked in the future-release backlog.

Document Structure

This protocol is organised so you can read it top-to-bottom for the full APPN plot delineation standard, or jump straight to the section you need. Decisions made at the **APPN Field EWG plot-delineation meeting** are reflected inline; outstanding follow-ups are flagged in the Outstanding TODOs at the top.

- APPN Plot Delineation — rationale and the competing sources of error a standard delineation approach must manage.
- Recommended Buffer — minimum inward buffer values, real-world worked examples, and when to deviate.
- APPN Plot Shapefile Standard — the mandatory file format, attributes, storage location, and naming convention every plot layout file must follow.
- File format — **GeoJSON ratified as the primary format**; shapefile retained as legacy / companion only.
- Required attributes — the ratified minimum set (`fid`, `plot_id`, `row`, `range`, `crop`) plus optional columns (`is_buffer`, `block`, etc.).
- Storage location — confirmed as `Documentation/Plot_Layout/`.
- File naming convention — sensor tag in the file name (e.g. `CALVIS`, `GOBI`, `HIRES`, `M3M`).
- Joining Trial Information — how trial metadata is attached to the plot geometry via `plot_id`.
- Methods — supported procedures for generating an APPN-compliant plot layout file.
- Method 1: FIELDimageR (QGIS)
- Method 2: DPIRD Field Mapping Tool
- Method 3: GPT plot creation tool

Important

Summary — the one mandatory plot file. Every APPN field site **must** have a single main plot file:

- **Filename:** `{YYYYSiteName}_plots.geojson`
- **Location:** `{Node}/{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/{YYYYSiteName[_F|C]}/Documentation/Plot_Layout/`
- **Example:** `USYD_Narrabri/2025_SIFOzBarley/2025IAWatson_F/Documentation/Plot_Layout/2025IAWatson_plots.geojson`

This file is the entry point every downstream pipeline depends on. It **must** carry the mandatory standard attributes described in Required attributes. All other files (`plots_*` variants, `sampling_*`, `gcp`) are optional. See Storage location and File naming convention for details.

APPN Plot Delineation

A standard APPN plot delineation approach ensures that comparable trials can be analysed consistently across nodes. The goal is to maximise the usable plot area sampled by the aerial data while minimising two competing sources of error:

- **Edge effects** — agronomic and radiometric contamination from neighbouring plots, alleys, and bare soil at plot boundaries.
- **Positional uncertainty** — small misalignments between the plot shapefile and the orthomosaic caused by GNSS/RTK error, orthorectification residuals, and sowing/layout drift (rows bowing or skewing relative to the design grid as the seeder tracks across the trial).

In practice, this is achieved by applying a consistent inward **buffer** to each plot polygon so the analysed region sits comfortably inside the true plot extent, regardless of which method is used to generate the shapefile.

Recommended Buffer

To keep results comparable across nodes, APPN trials should apply at least the inward buffer specified in the table below to every plot polygon. The current working rule is:

Important

APPN minimum buffer (working rule): 0.3 m from each plot end and 0.2 m from each plot side (across the drill direction), **or** the width of one row — whichever is larger.

Note

The "one row width" fallback is intended to handle **sorghum and other widely-spaced crops**, where the fixed 0.2 m side buffer is narrower than a single inter-row spacing and would still capture neighbouring rows. A more general species-specific rule is still pending with the EWG.

The buffer used must be recorded in the tool-specific configuration saved alongside the shapefile (e.g. the FIELD-imageR JSON) so the layout can be reproduced.

Worked examples (real-world APPN plot sizes)

Plot size (L × W)	Min. buffer (end / side)	Analysis area (L × W)	% of plot
6 m × 2 m (common cereal yield plot)	0.3 m / 0.2 m	5.4 m × 1.6 m	~72 %
6 m × 1.5 m (DPIRD standard)	0.3 m / 0.2 m	5.4 m × 1.1 m	~66 %
4 m × 1.5 m (UOA OzBarley)	0.3 m / 0.2 m	3.4 m × 1.1 m	~62 %
10 m × 3 m (Sorghum agronomy strip)	1.0 m / 0.5 m	8.0 m × 2.0 m	~53 %

> **Note**
>
> Plot widths can vary substantially between trials (Ingrid — UOA), so the table is **illustrative**, not exhaustive. Apply the working rule above to any plot size not listed.

When to increase the buffer

- Coarser GSD (e.g. hyperspectral at ~5 cm vs RGB at ~1 cm).
- Tall or lodging-prone canopies where canopy lean shifts the visible plot off its sown footprint.
- Narrow alleys (<0.5 m) where neighbouring canopies merge.
- Trials without RTK GNSS or without ground control points (GCPs) in the orthomosaic.

> **Note**
>
> Any deviation from the default buffer must be recorded with the trial's plot layout files and justified in the trial notes.

APPN Plot Shapefile Standard

All APPN plot shapefiles must conform to the following standard so that downstream pipelines can ingest them without trial-specific configuration.

File format

- **Primary format: GeoJSON** (`.geojson`) — a single, plain-text, self-contained file. See File Format — GeoJSON vs Shapefile in the Aerial Data QC protocol for the full rationale (single-file packaging, version-control friendliness, no field-name length or file size caps, open web-native standard).
- **Legacy / companion format:** ESRI Shapefile (`.shp` plus its sidecar files `.shx`, `.dbf`, `.prj`, `.cpg`). All sidecar files must be kept together with the `.shp`. Existing shapefiles do **not** need to be re-created, but **all new APPN files must be saved as `.geojson`**.
- **CRS:** the CRS of the source orthomosaic (typically the correct zone of GDA2020). For GeoJSON, keep the file in the projected CRS of the orthomosaic rather than reprojecting to WGS84 (see the linked rationale above). For shapefiles, the `.prj` file must be present and correct.
- **Geometry:** one polygon per plot. Polygons should be rectangular and aligned to the trial layout, sized to the plot dimensions minus the inward buffer applied to mitigate edge effects.
- **Optional companion copy:** a second copy of the same layer in shapefile form may be saved alongside the primary GeoJSON using the **same base file name** (e.g. `MyTrial_plots.geojson` → `MyTrial_plots.shp`). This is useful for tools that only consume shapefiles, but is not required.

Required attributes

> **Important**
>
> **Field EWG decision:** the **mandatory attribute set is now ratified** — see the table below. `species` has been **removed**. Optional columns must be **omitted entirely** when no data is available (rather than carried as empty / placeholder columns).

Every APPN plot polygon **must** carry the following attributes:

Column	Required?	Description
`fid`	Mandatory	Unique polygon identifier assigned by the delineation tool. Identifies the *geometry* only and may not match the trial's plot numbering.
`plot_id`	Mandatory	Plot number from the trial design / sowing plan. **Join key** for trial metadata (see Joining Trial Information).
`row`	Mandatory	Row index in the trial design.
`range`	Mandatory	Range (column) index in the trial design.
`crop`	Mandatory	Crop type (e.g. *Wheat*).

Note

`fid` and `plot_id` must be kept as **separate columns**. `fid` is the tool's internal polygon ID; `plot_id` is the agronomic plot number from the trial design. Conflating the two breaks reproducibility when the shapefile is regenerated and `fid` values shift.

`plot_id` (not `fid`) is the join key for trial metadata. Trial information arrives in a wide range of formats and structures across nodes and projects and cannot realistically be standardised into a single schema, so at a minimum the shapefile and the trial spreadsheet must share `plot_id` for the join described in Joining Trial Information to be performed reliably. `fid` must not be used as the join key — it is tool-assigned and may change when the shapefile is regenerated.

Optional attributes The following columns are **optional**. If the data is no availble they should be **omitted entirely**.

Trial design:

- `is_buffer` — Boolean (`True`/`False`) flag marking *buffer plots* (filler / border plots that absorb edge effects and carry no experimental treatment). Not to be confused with the inward analysis **buffer** applied to every plot polygon (see Recommended Buffer).
- `block` — replication block / replicate identifier from the trial design. Include whenever the trial design defines blocks.

Biological / treatment:

- `genotype` / `entry` — variety, line, or accession code. If this information is proprietary, an anonymised code may be used instead.
- `treatment` — agronomic or experimental treatment applied to the plot.

Provenance (used to trace how the polygon was produced):

- `method` — delineation method (e.g. `FIELDimageR`, `DPIRD`, `GPT`).
- `buffer_end_m`, `buffer_side_m` — buffer values applied (in metres).
- `source_file` — filename or ID of the orthomosaic the polygon was fit to.
- `created` — ISO date the file was generated.
- `created_by` — Name of the person who produced the file, for provenance and follow-up queries.
- `notes` — Any additional infomation about creation or usage

Note

Both Method 1 (FIELDimageR) and Method 2 (DPIRD Field Mapping Tool) have their own metadata-capture conventions, and harmonising these into a single standard is still outstanding. Metadata should be saved alongside the layer file. Where additional metadata needs to be added manually, use the schema below.


```
{
 "type": "FeatureCollection",
 "name": "2025IAWatson_plots_v01",
 "crs": {
 "type": "name",
 "properties": { "name": "urn:ogc:def:crs:EPSG::7855" }
 },
 "provenance_metadata": {
 "datasetId": "2025IAWatson_plots_v01",
 "site": "2025IAWatson",
 "node": "USYD_Narrabri",
 "method": "FIELDimageR",
 "source_file": "2025IAWatson_CALVIS_Orthomosaic.tif",
 "sensor": "CALVIS",
 "buffer_end_m": 0.3,
 "buffer_side_m": 0.2,
 "created": "2026-05-18",
 "created_by": "Arden Burrell",
```

```
 "originatingSystem": "QGIS 3.36 + FIELDimageR",
 "notes": "Buffer applied per APPN minimum working rule."
 },
 "features": [
 {
 "type": "Feature",
 "geometry": {
 "type": "Polygon",
 "coordinates": [ [ [ /* … */ ] ] ]
 },
 "properties": {
 "fid": 1,
 "plot_id": 1001,
 "row": 1,
 "range": 1,
 "crop": "Wheat",
 "is_buffer": false,
 "block": 1,
 "genotype": "Scepter",
 "treatment": "N0"
 }
 }
 /* … one Feature per plot … */
 ]
}
```

> **Note**
>
> `crs` should match the projected CRS of the source orthomosaic (typically the relevant GDA2020 / MGA zone — `EPSG:7855` is shown here for Narrabri NSW). `provenance_metadata` is a top-level sidecar object that records how the layer was produced and mirrors the per-feature provenance attributes listed above. Per-feature `properties` must include the Required attributes and may include any of the optional attributes when the data is available.

Storage location

Save the plot layout file (GeoJSON, or shapefile with all its sidecar files) in the site-level `Documentation/Plot_Layout/` directory under the APPN folder structure (see the APPN folder structure wiki for the full naming convention).

Formal path:

```
{Node}/
 {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
 {YYYYSiteName[_F|C]}/Documentation/Plot_Layout/
```

Example:

```
USYD_Narrabri/2025_SIFOzBarley/2025IAWatson_F/Documentation/Plot_Layout/
```

Also save the tool-specific configuration used to generate the layout file (e.g. the FIELDimageR JSON settings) alongside it so the layout can be reproduced.

Folder README

Each `Plot_Layout/` folder **should** contain a `README.md` capturing site-specific context that doesn't fit cleanly into per-file `provenance_metadata`: which file is the current main `plots` layout, why any variant (`plots_unbuffered`, `plots_rszone`, `plots_{sensor}`) exists, a deprecation log, sampling-campaign notes, and any known quirks. Machine-readable provenance still lives in each file's `provenance_metadata`; the README is the human-readable companion for the folder as a whole.

Suggested template:

```
# Plot Layout - {YYYYSiteName}

Site-specific notes for the files in this folder. For the APPN-wide
spec see the [Plot Delineation protocol](…/Plot_Delineation.md).

## Current main file
- `{YYYYSiteName}_plots.geojson` - fitted YYYY-MM-DD by <name>,
 method <FIELDimageR | DPIRD | GPT>.

## Variants in use
- `plots_unbuffered` - used by <pipeline / person> for <reason>.
- `plots_{sensor}` - justified because <CRS / portion-of-plot reason>;
 approved by EWG on <date>.

## Sampling campaigns
- `sampling_biomass_YYYYMMDD…` - operator, quadrat size, anything odd.

## Deprecated files
| File | Replaced on | Reason | Superseded by |
| --- | --- | --- | --- |
| `…_plots_20250901_deprecated.geojson` | 2025-09-01 | Re-fit after RTK base correction |
`…_plots.geojson` |

## Known issues / quirks
- e.g. "HIRES flight 2025-10-12 had a 7 cm N-S offset - corrected in
 re-process."
```

File naming convention

A site's `Plot_Layout/` directory holds **three families** of vector file, and the naming convention is built around that split:

1. **Plot files (`plots*`)** — define the **plot polygons** (the area to analyse). Every site has **one main `plots` file** with a short, simple name that is reused across every sensor wherever possible. Additional plot files may be added with longer, descriptive suffixes that record their more limited usage — for example, an unbuffered variant, a sensor-specific variant, or a remote-sensing- only sub-area.
2. **Sampling files (`sampling_*`)** — record the **portion of each plot that was actually sampled** by a destructive or in-field measurement (biomass cut, height measurement, emergence count, head count, etc.). One file per measurement type, named after the measurement.
3. **GCP file (`gcp`)** — **permanent ground control points only** (fixed, surveyed markers that persist across flights and seasons). Temporary or per-flight GCPs are **not** stored here — they belong with the flight that used them.

> **Important**
>
> **The main `plots` file is the only mandatory vector file.** Every APPN project and site **must** have a `{YYYYSiteName}_plots.geojson` in its `Plot_Layout/` folder — it is the entry point every downstream pipeline depends on. All other files in the three families (`plots_*` variants, `sampling_*`, `gcp`) are **optional** and are added only when the trial actually needs them. A `README.md` in the folder (see Folder README) is **strongly recommended** for every site, even when only the main `plots` file is present.

Working format Three parallel patterns — one per family (`plots`, `sampling`, and `gcp`):

```
{YYYYSiteName}_plots[_{descriptor}][_{sensor}][_{YYYYMMDD}][_v{NN}][_deprecated].{ext} # plot files
{YYYYSiteName}_sampling_{measurement}[_{YYYYMMDD}][_v{NN}][_deprecated].{ext} # sampling files
{YYYYSiteName}_gcp[_{YYYYMMDD}][_v{NN}][_deprecated].{ext} # ground control points
```

Common fields (apply to all three patterns):

Field	Notes
`{YYYYSiteName}`	Site identifier (year + site name), matching the parent folder name with the `_F` suffix dropped. Plot layouts only apply to field sites.
`{YYYYMMDD}`	**Optional** ISO-style date tag (e.g. `20251104`) identifying the date the file applies to — typically the **sampling / measurement date** on `sampling_*` files, or the **survey date** on a permanent `gcp` file. Omit on the main `plots` file (which spans the whole season). **Mandatory** on `sampling_*` files when more than one event of the same measurement occurs in a season.**Multi-date / range syntax:** when a single sampling event spans more than one day, combine dates in the tag itself — use `and` to list discrete dates (`20251104and20251125`) and `to` for a continuous date range (`20251104to20251108`). The two joiners may be combined (`20251104to20251108and20251215`). Keep dates in chronological order.
`_v{NN}`	Optional zero-padded revision (`_v01`, `_v02`, …). Bump on any change to geometry or attributes.
`_-deprecated`	**Optional** terminal tag flagging a file that has been **superseded but retained for provenance** (e.g. the older revision of a re-fit plot layout, or a measurement file replaced by a corrected version). The replacement file keeps the original name (or a bumped `_v{NN}`); the superseded file is renamed with this suffix appended so downstream pipelines skip it. Record the reason for deprecation and the name of the superseding file in the deprecated file's `provenance_metadata.notes`.
`{ext}`	`geojson` (mandated primary). `shp` (with sidecars) is also accepted as a legacy / companion file.

Plot-file fields (`plots[_{descriptor}][_{sensor}]`):

Field	Notes
`{descriptor}`	**Optional** short tag for a non-default plot variant (`unbuffered`, `rszone`, …). Omit on the main file.
`{sensor}`	**Optional** sensor tag — use the platform name (`CALVIS`, `GOBI`, `HIRES`, `M3M`). Only add a sensor-specific file when the geometry genuinely differs — see the 5 cm rule above.

Sampling-file fields (`sampling_{measurement}`):

Field	Notes
`{measurement}`	**Mandatory** on sampling files. Short tag identifying the measurement type (`biomass`, `height`, `emergence`, `headcount`, …).

Plot file tags

- `plots` — **the main file.** Primary analysis layout (one polygon per plot, buffer applied per the APPN Plot Shapefile Standard). Used by every sensor unless a sensor-specific variant is justified.

> **Note**
>
> The main `plots` file is the entry point every downstream pipeline looks for, so it should keep its **short, unversioned name** (`{YYYYSiteName}_plots.geojson`) for the life of the trial. When the layout is re-fit and a new version supersedes the current main file, **prefer the `_deprecated` tag over bumping `_v{NN}` on the main file** — rename the old file to `{YYYYSiteName}_plots_{YYYYMMDD}_deprecated.geojson` (date tag disambiguates multiple deprecations) and save the new layout back under the plain `{YYYYSiteName}_plots.geojson` name. This keeps the main file easy to find and unambiguous, and keeps older copies clearly marked as superseded.

- `plots_unbuffered` — full plot footprint with **no inward analysis buffer**, for cases where the entire plot area is needed.
- `plots_rszone` — sub-area within each plot **reserved for remote sensing** and kept off-limits to destructive sampling.
- `plots_{sensor}` — sensor-specific geometry variant, where `{sensor}` is the platform name (`CALVIS`, `GOBI`, `HIRES`, or `M3M`). Only create one when the sensor geometry genuinely differs (see 5 cm rule above).

> **Important**
>
> **When a `plots_{sensor}` file is justified.** A sensor-specific plot file should only be created when the sensor *systematically* needs different polygons from the main `plots` file. The two accepted justifications are:
>
> 1. **The sensor exports in a different CRS** and cannot be reprojected losslessly into the standard GDA2020 / MGA zone of the main `plots` file.
> 2. **The sensor captures a different portion of the plot** (e.g. `Omega` capturing the entire plot without bufffer), so a single shared polygon would systematically clip or over-extend the analysis area.
>
> **Hard rule:** if plot-file differences **within a single sensor** exceed ~**5 cm**, do **not** create extra files to work around them — escalate and address the root cause (georeferencing, orthorectification, GCPs).

Strongly discouraged: stacking further tags such as `plots_{sensor}_{YYYYMMDD}` — or, worse, `plots_{sensor}_{YYYYMMDD}_{run}` — to paper over a single bad run (e.g. one flight with the wrong CRS or a spatial misalignment). This fragments the plot layout across files, breaks downstream pipelines that key on the main `plots` file, and hides a real data-capture problem. Re-process the offending run instead. Per-run overrides at this level are a **last resort** — use only after all other attempts to fix the data have failed, and record the justification in the file's `provenance_metadata.notes`.

Sampling file tags One file per measurement type. Extensible — add new tags as new measurements are introduced.

- `sampling_biomass` — footprints of biomass cuts within each plot. Whole-plot biomass collection (e.g. UOA all-of-plot workflow) uses the same `sampling_biomass` tag with polygons matching the full plot extent — differentiate it via the date tag, not a separate filename suffix.
- `sampling_height` — locations / quadrats where canopy height was measured.
- `sampling_emergence` — emergence-count quadrats.
- `sampling_headcount` — headcount quadrats.
- `sampling_{measurement}` — extensible for other destructive or in-field sampling (manual quadrats, soil cores, damage assessments, etc.).

Other tags

- `gcp` — **permanent** ground control point locations only (fixed, surveyed markers that persist across flights and seasons). Temporary or per-flight GCPs must **not** be stored here — they belong with the flight that used them.

Examples Within `USYD_Narrabri/2025_SIFOzBarley/2025IAWatson_F/Documentation/Plot_Layout/`:

```
# Plot files
2025IAWatson_plots.geojson (main file - used by every sensor)
2025IAWatson_plots_unbuffered_v01.geojson (full plot footprint, no inward buffer)
2025IAWatson_plots_rszone_v01.geojson (RS-safe sub-area, no destructive sampling)
2025IAWatson_plots_HIRES_v01.geojson (sensor variant - only if geometry differs)
2025IAWatson_plots_v01.shp (+ .shx .dbf .prj .cpg - optional companion)
2025IAWatson_plots_v01.json (FIELDimageR settings sidecar)

# Sampling files (one per measurement type; date tag identifies the sampling event)
2025IAWatson_sampling_biomass_20251104_v01.geojson (single biomass cut, 2025-11-04)
2025IAWatson_sampling_biomass_20251104and20251125_v01.geojson (biomass cut spread over two discrete days)
2025IAWatson_sampling_biomass_20251201to20251205_v01.geojson (biomass cut over a 5-day window)
2025IAWatson_sampling_emergence_20250710_v01.geojson
2025IAWatson_sampling_headcount_20251018_v01.geojson

# Ground control points
2025IAWatson_gcp_v01.geojson

# Superseded but retained for provenance
2025IAWatson_plots_20250901_deprecated.geojson (previous main layout, replaced 2025-09-01 -
see notes)
```

Note

All `sampling_*` and `plots_*` variant files should use the **same CRS and `plot_id` scheme** as the main `plots` file so that downstream pipelines can spatially join, subtract, or flag affected plots without additional configuration. The **only** accepted exception is when a sensor cannot output in the standard GDA2020 CRS — in that case the variant file may use the sensor's native CRS, but the CRS must be declared in the file's `provenance_metadata` and a reprojection step added to the downstream pipeline.

Joining Trial Information

Trial information arrives in a wide range of formats and structures across nodes, projects, and collaborators (trial designs, agronomy records, breeder spreadsheets, LIMS exports) and **cannot realistically be standardised** into a single APPN schema. Instead of mandating a spreadsheet layout, APPN mandates **one thing only**: the trial spreadsheet must carry a `plot_id` column whose values match the `plot_id` attribute of the plot GeoJSON exactly (same type, same zero-padding, no whitespace). Everything else is up to the trial.

> **Important**
>
> **The only hard requirement is the join key.** The trial spreadsheet **must** include a `plot_id` column that matches the plot file's `plot_id` one-to-one. `fid` must **not** be used as the join key — it is tool-assigned and changes when the plot file is regenerated.

Suggested columns

The columns below are **examples**, not a required schema. Include the ones the trial actually uses; omit the rest. Add any trial-specific columns you need.

Column	Typical use
`plot_id`	**Required.** Join key — must match the plot file.
`row`, `range`	Trial-design coordinates (handy for sanity-checking the join).
`block` / `replicate`	Replication block from the trial design.
`crop`	Crop type (e.g. `Wheat`).
`genotype` / `entry`	Variety, line, or accession code (anonymised if proprietary).
`treatment`	Agronomic / experimental treatment (e.g. `N0`, `Irrigated`).
`sowing_date`	ISO date sown.
`seed_rate`, `row_spacing_m`	Agronomy parameters when relevant.
`notes`	Free-text per-plot notes (damage, missing rows, etc.).

Example template (`{YYYYSiteName}_trial_info.csv`)

```
plot_id,row,range,block,crop,genotype,treatment,sowing_date,notes
1001,1,1,1,Wheat,Scepter,N0,2025-05-12,
1002,1,2,1,Wheat,Mace,N0,2025-05-12,
1003,1,3,1,Wheat,Scepter,N100,2025-05-12,
2001,2,1,2,Wheat,Mace,N100,2025-05-12,missing first row
...
```

CSV is preferred (plain text, diff-friendly, opens everywhere); XLSX is acceptable if the trial team already maintains it.

Storage location

Trial-information files live in a dedicated **`Trial_Info/`** subfolder alongside `Plot_Layout/` under the site's `Documentation/` directory:

```
{Node}/
 {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
 {YYYYSiteName[_F|C]}/Documentation/
 Plot_Layout/
 Trial_Info/
```

Example:

```
USYD_Narrabri/2025_SIFOzBarley/2025IAWatson_F/Documentation/Trial_Info/
 2025IAWatson_trial_info.csv (current trial info)
 2025IAWatson_trial_info_20250612_deprecated.csv (superseded version)
 README.md (source spreadsheets, contacts, quirks)
```

The same `_deprecated` and `_v{NN}` conventions used for plot files apply here (see File naming convention). Record the current filename and any quirks in the folder's `README.md`, and reference it from the `Plot_Layout/` Folder README so the two folders stay linked.

Methods

The following methods are supported for generating an APPN-compliant plot shapefile. Choose the method that best matches your imagery and tooling; the resulting shapefile must satisfy the APPN Plot Shapefile Standard above.

- Method 1: FIELDimageR (QGIS)
- Method 2: DPIRD Field Mapping Tool
- Method 3: GPT plot creation tool — GRYFN Plot Extraction Tool workflow (replaces the previous "GRYFN plot tool" placeholder).

Method 1: FIELDimageR (QGIS)

FIELDimageR is an R-based plugin run from within QGIS that builds plot polygons from a georeferenced orthomosaic.

Software Installation

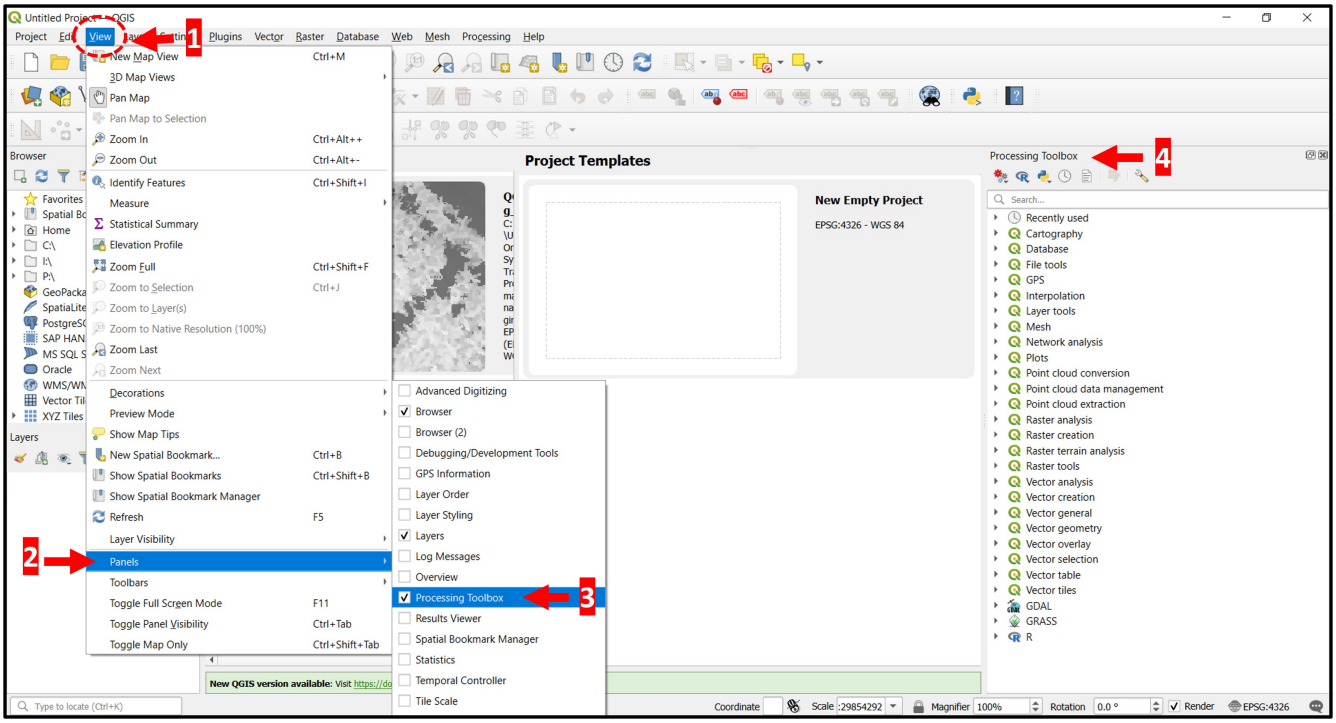
Install the following software to start the pipeline:

1. R
2. QGIS

> **Note**
>
> The first time you run FIELDimageR-QGIS it may take some time to install all required R packages.

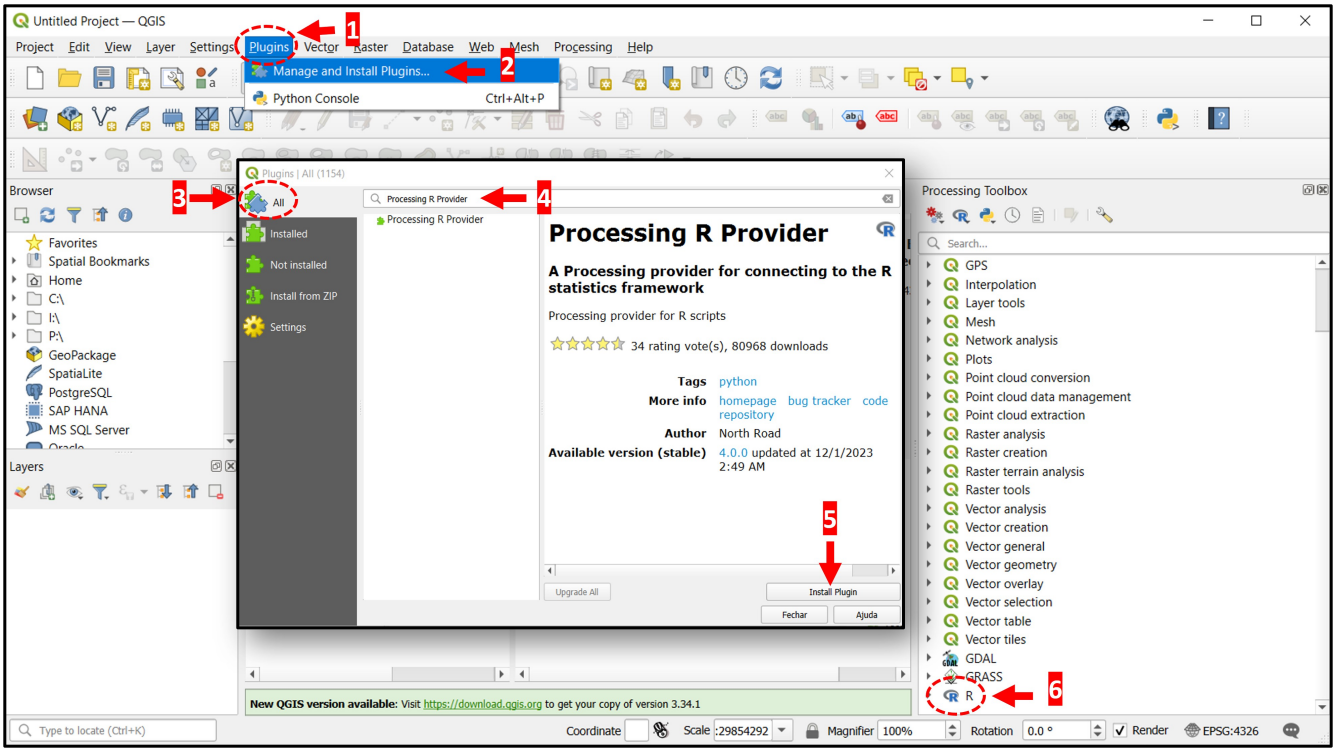
Enable the Processing Toolbox in QGIS Make sure the **Processing Toolbox** panel is visible in QGIS:

1. Open the **View** menu.
2. Select **Panels**.
3. Enable **Processing Toolbox**.
4. Confirm the Processing Toolbox is now showing on the right-hand side.



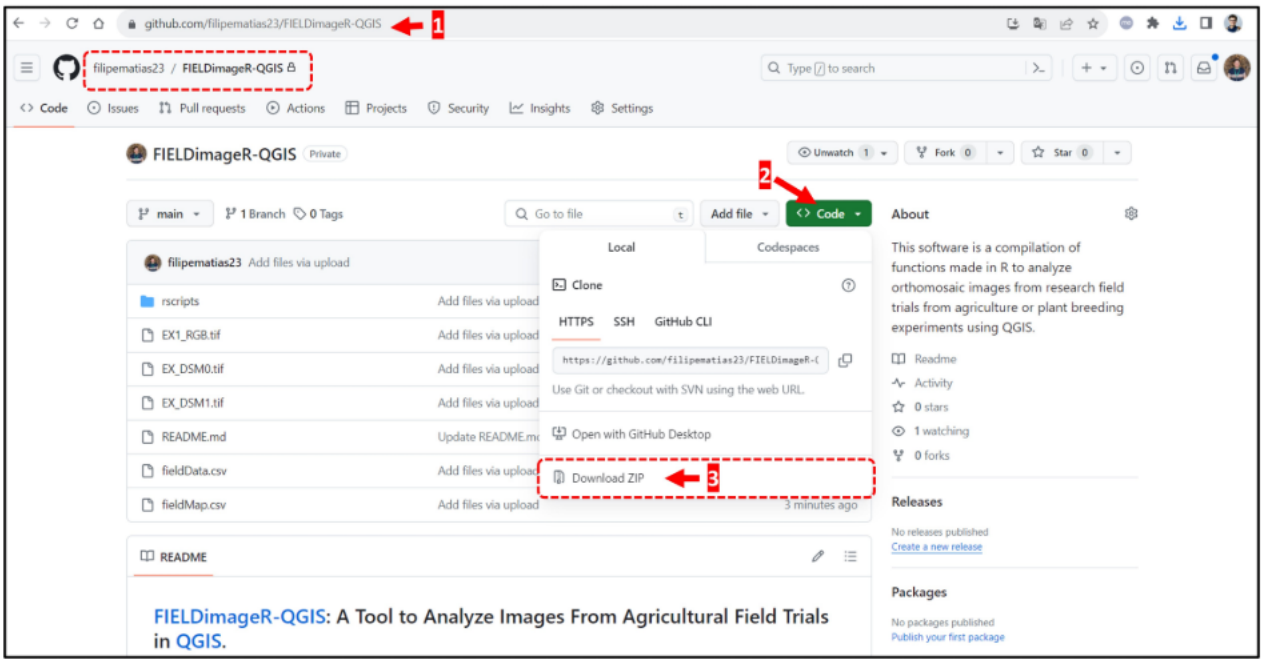
Install the Processing R Provider plugin Install the **Processing R Provider** plugin in QGIS:

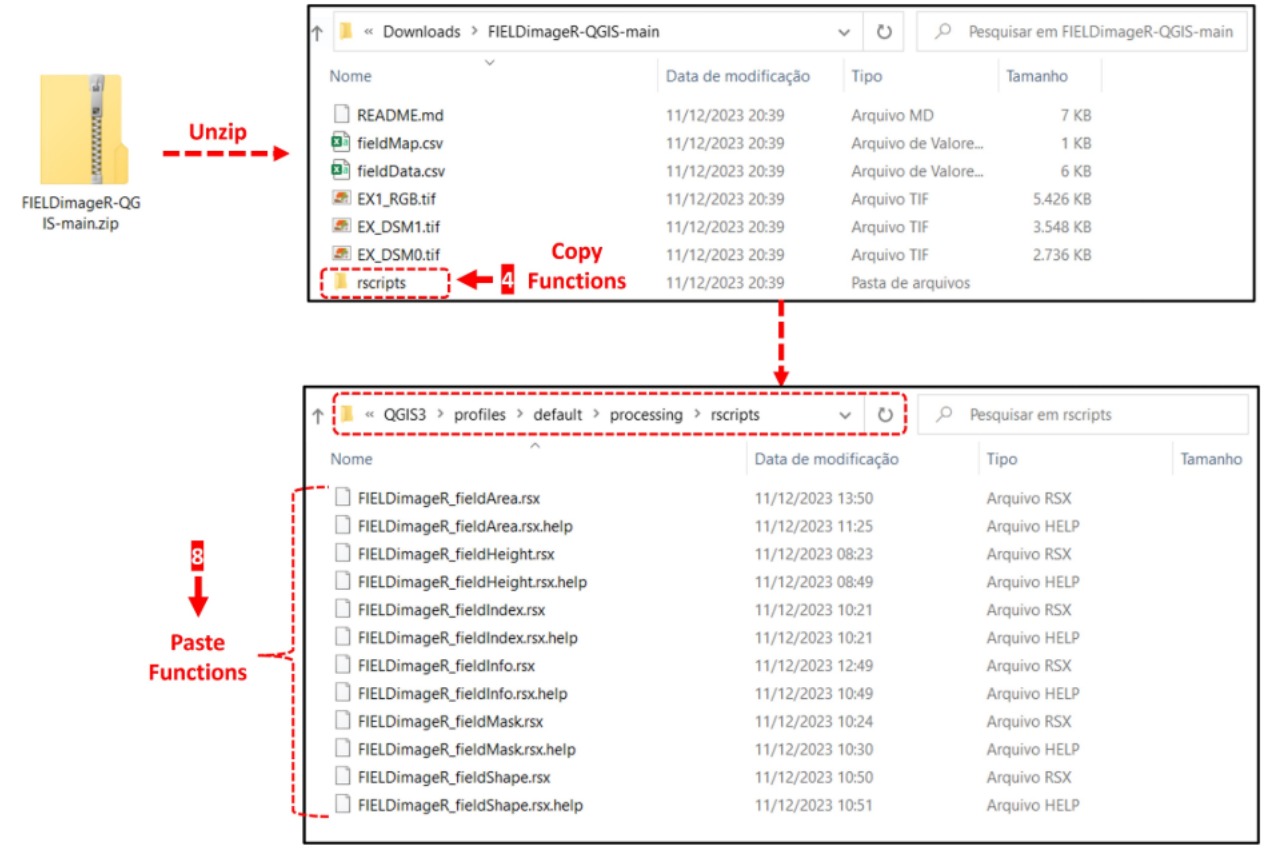
1. Open the **Plugins** menu.
2. Select **Manage and Install Plugins**.
3. Switch to the **All** tab.
4. Search for *Processing R Provider*.
5. Click **Install Plugin**.
6. Verify that **R** now appears in the Processing Toolbox.

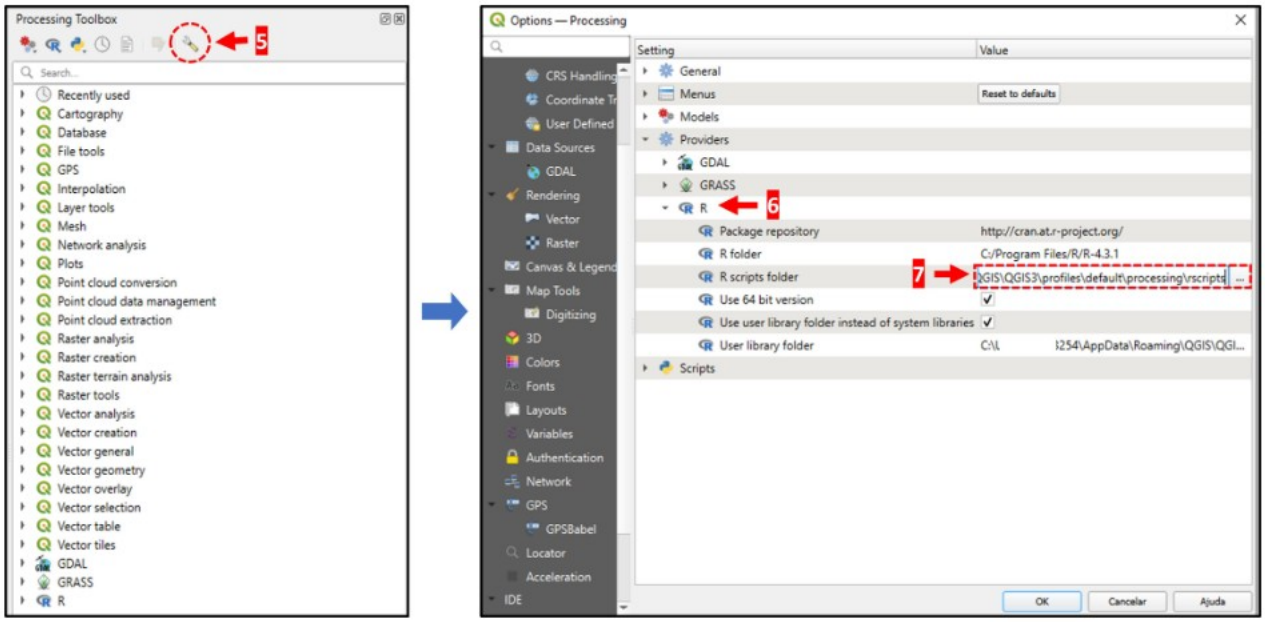


Install FIELDimageR

1. Go to the FIELDimageR-QGIS GitHub repository: https://github.com/filipematias23/FIELDimageR-QGIS.
2. Click **Code**.
3. Select **Download ZIP**.
4. Unzip the archive and copy the functions from the `rscripts` folder into your **QGIS R scripts** folder.
5. To locate the QGIS R scripts folder, go to **QGIS** → **Processing Toolbox** → **Options** (the wrench icon).
6. Under **Providers**, click **R**.
7. Copy the path shown for **R scripts folder** and open it in your file explorer.
8. Paste the FIELDimageR functions downloaded from GitHub into the `rscripts` folder.


github.com/filipematias23/FIELDimageR-QGIS
1
filipematias23 / FIELDimageR-QGIS
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
FIELDimageR-QGIS
Private
2
main
1 Branch
0 Tags
Go to file
Add file
Code
About
filipematias23 Add files via upload
rscripts
EX1_RGB.tif
EX_DSM0.tif
EX_DSM1.tif
README.md
fieldData.csv
fieldMap.csv
Add files via upload
Update README.md
3 minutes ago
Local
Codespaces
Clone
HTTPS
SSH
GitHub CLI
Use Git or checkout with SVN using the web URL.
Open with GitHub Desktop
Download ZIP
3
This software is a compilation of functions made in R to analyze orthomosaic images from research field trials from agriculture or plant breeding experiments using QGIS.
Readme
Activity
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
README
FIELDimageR-QGIS: A Tool to Analyze Images From Agricultural Field Trials in QGIS.

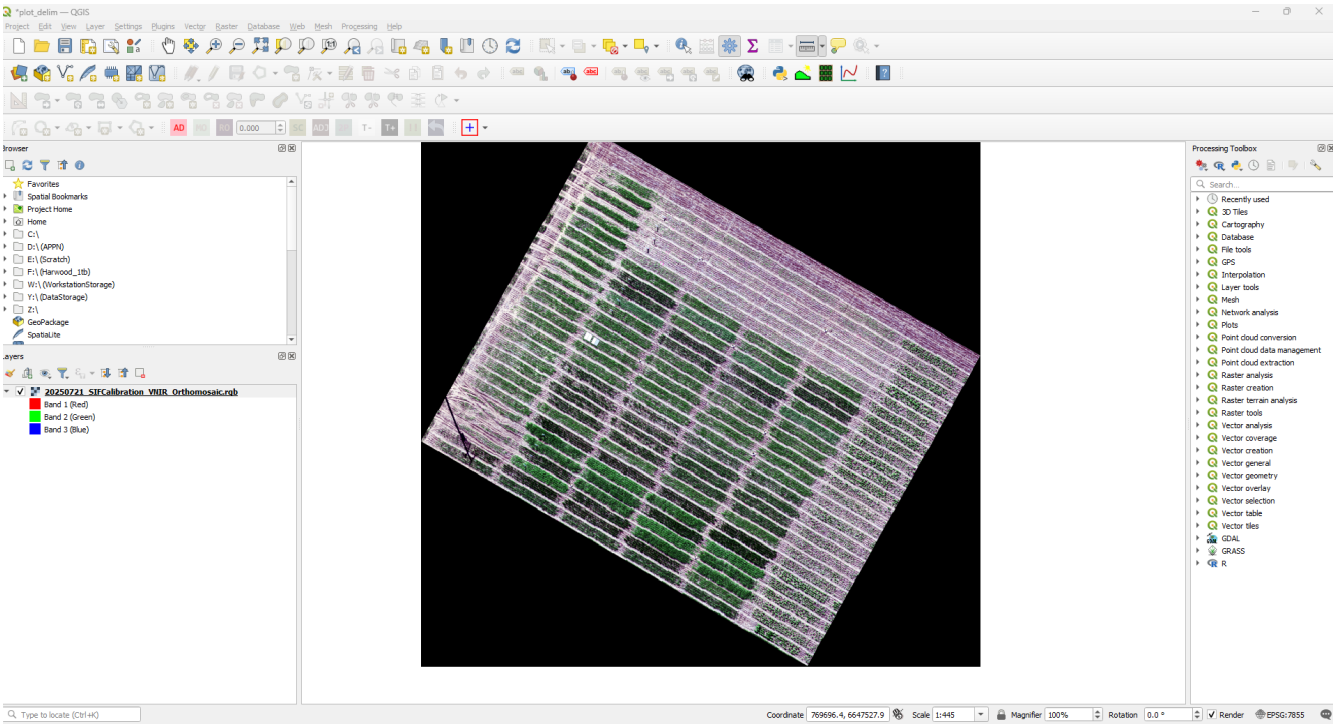

FIELDimageR-QGIS-main.zip
Unzip
Downloads
FIELDimageR-QGIS-main
Pesquisar em FIELDimageR-QGIS-main
Nome
Data de modificação
Tipo
Tamanho
README.md 11/12/2023 20:39 Arquivo MD 7 KB
fieldMap.csv 11/12/2023 20:39 Arquivo de Valore... 1 KB
fieldData.csv 11/12/2023 20:39 Arquivo de Valore... 6 KB
EX1_RGB.tif 11/12/2023 20:39 Arquivo TIF 5.426 KB
EX_DSM1.tif 11/12/2023 20:39 Arquivo TIF 3.548 KB
EX_DSM0.tif 11/12/2023 20:39 Arquivo TIF 2.736 KB
rscripts 11/12/2023 20:39 Pasta de arquivos
4
Copy Functions
8
Paste Functions
QGIS3
profiles
default
processing
rscripts
Pesquisar em rscripts
FIELDimageR_fieldArea.rsx 11/12/2023 13:50 Arquivo RSX
FIELDimageR_fieldArea.rsx.help 11/12/2023 11:25 Arquivo HELP
FIELDimageR_fieldHeight.rsx 11/12/2023 08:23 Arquivo RSX
FIELDimageR_fieldHeight.rsx.help 11/12/2023 08:49 Arquivo HELP
FIELDimageR_fieldIndex.rsx 11/12/2023 10:21 Arquivo RSX
FIELDimageR_fieldIndex.rsx.help 11/12/2023 10:21 Arquivo HELP
FIELDimageR_fieldInfo.rsx 11/12/2023 12:49 Arquivo RSX
FIELDimageR_fieldInfo.rsx.help 11/12/2023 10:49 Arquivo HELP
FIELDimageR_fieldMask.rsx 11/12/2023 10:24 Arquivo RSX
FIELDimageR_fieldMask.rsx.help 11/12/2023 10:30 Arquivo HELP
FIELDimageR_fieldShape.rsx 11/12/2023 10:50 Arquivo RSX
FIELDimageR_fieldShape.rsx.help 11/12/2023 10:51 Arquivo HELP



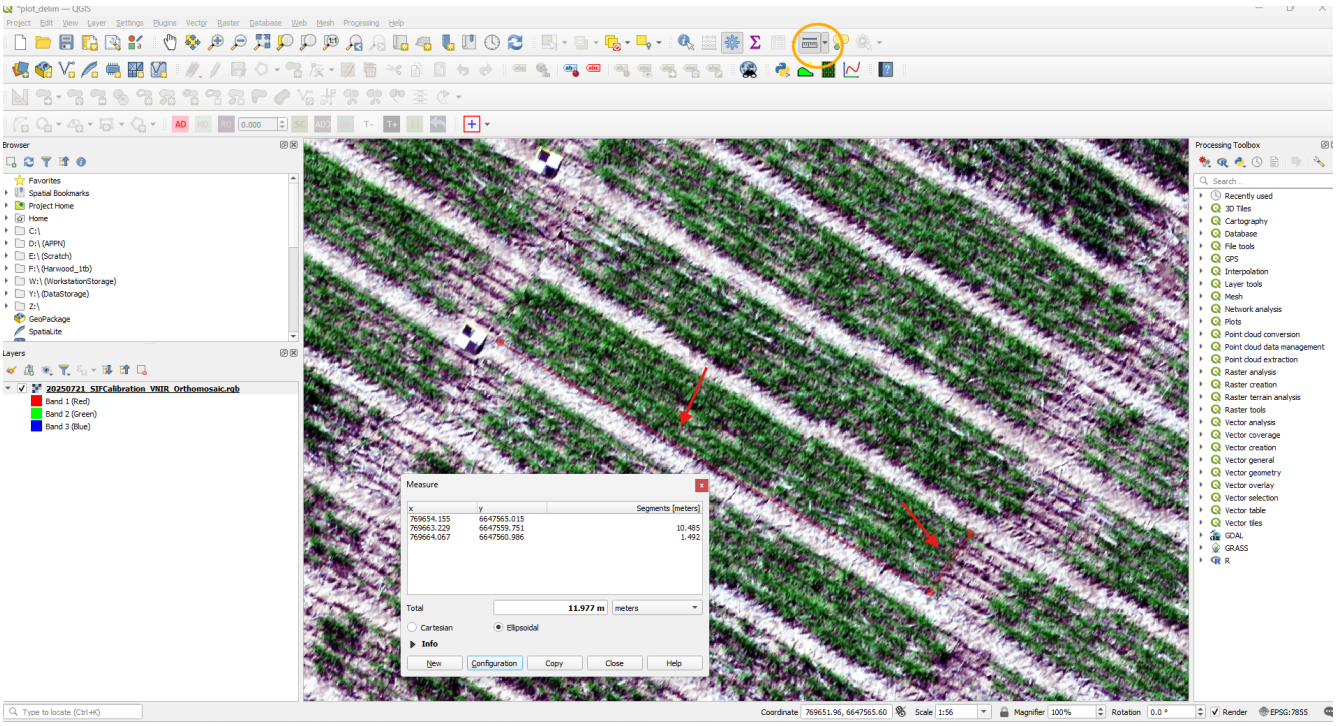
Generating the Plot Shapefile

With FIELDimageR set up, you can now generate plot shapefiles from your aerial imagery.

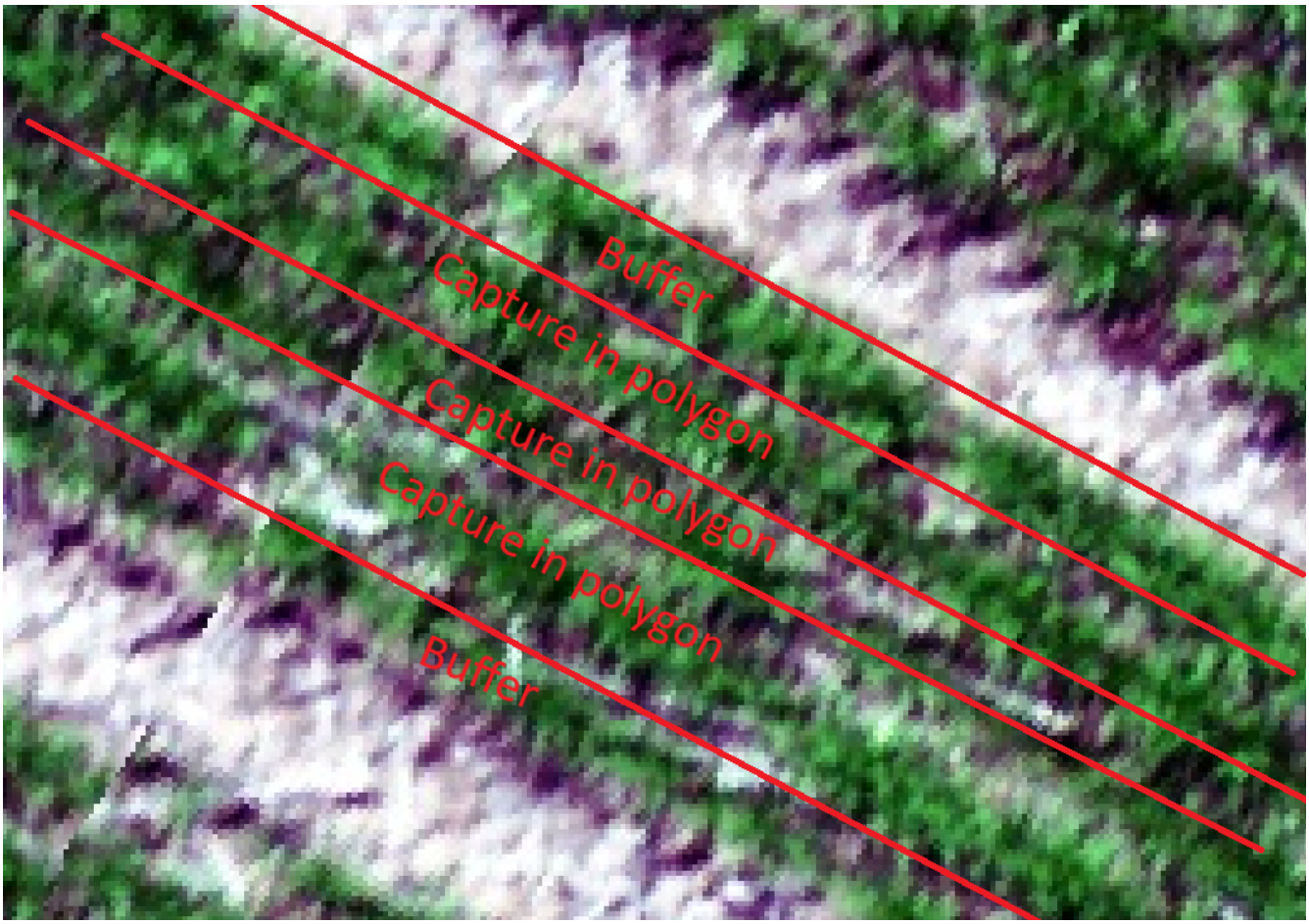
1. Load an image into QGIS. An RGB orthomosaic from a GRYFN drone is the easiest starting point.



Now it's important to inspect the plots and think about the plot boundaries. If we zoom into a plot and measure the width and the length we see the plot is ~ 10 meters by 1.5 meters.



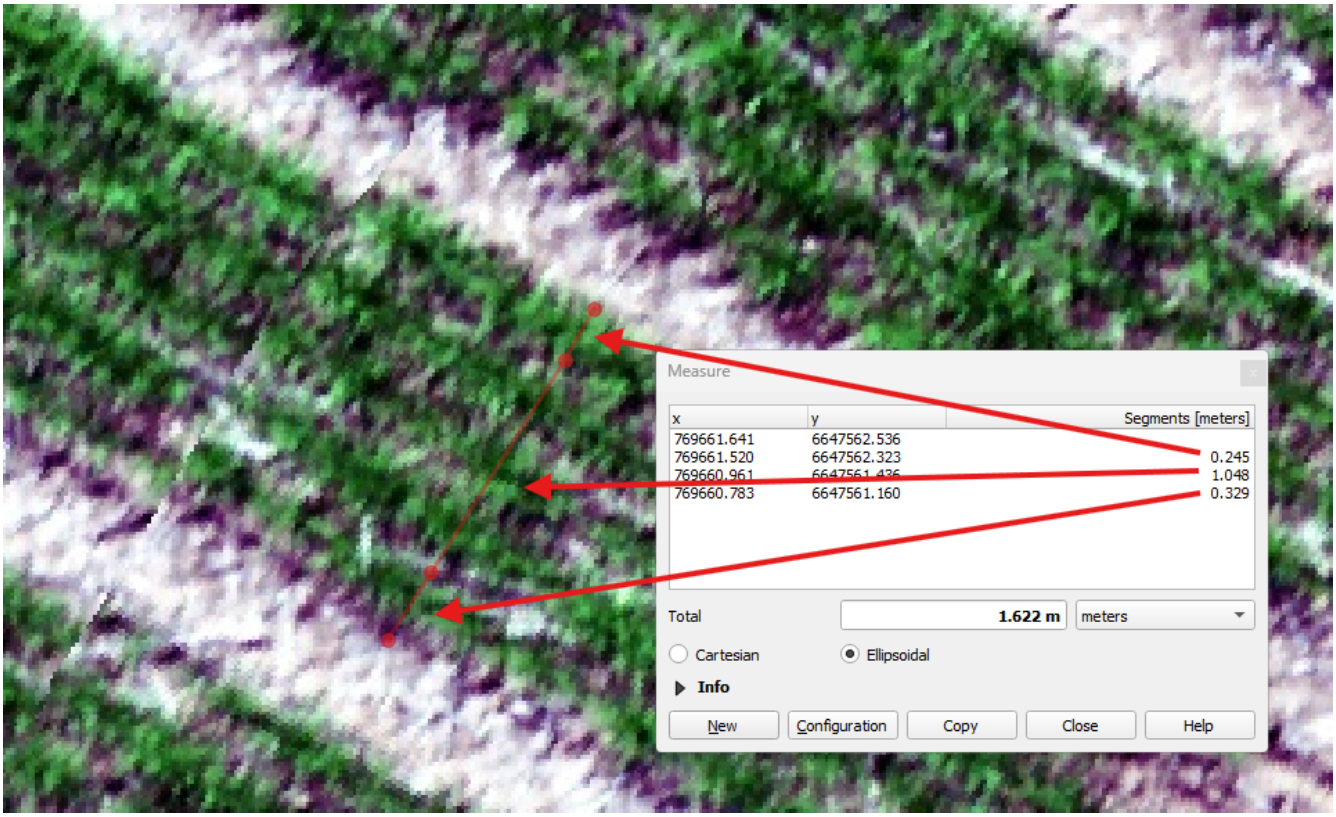
It has 5 rows (2 buffer rows, 3 rows we want to capture in our plant boundary)



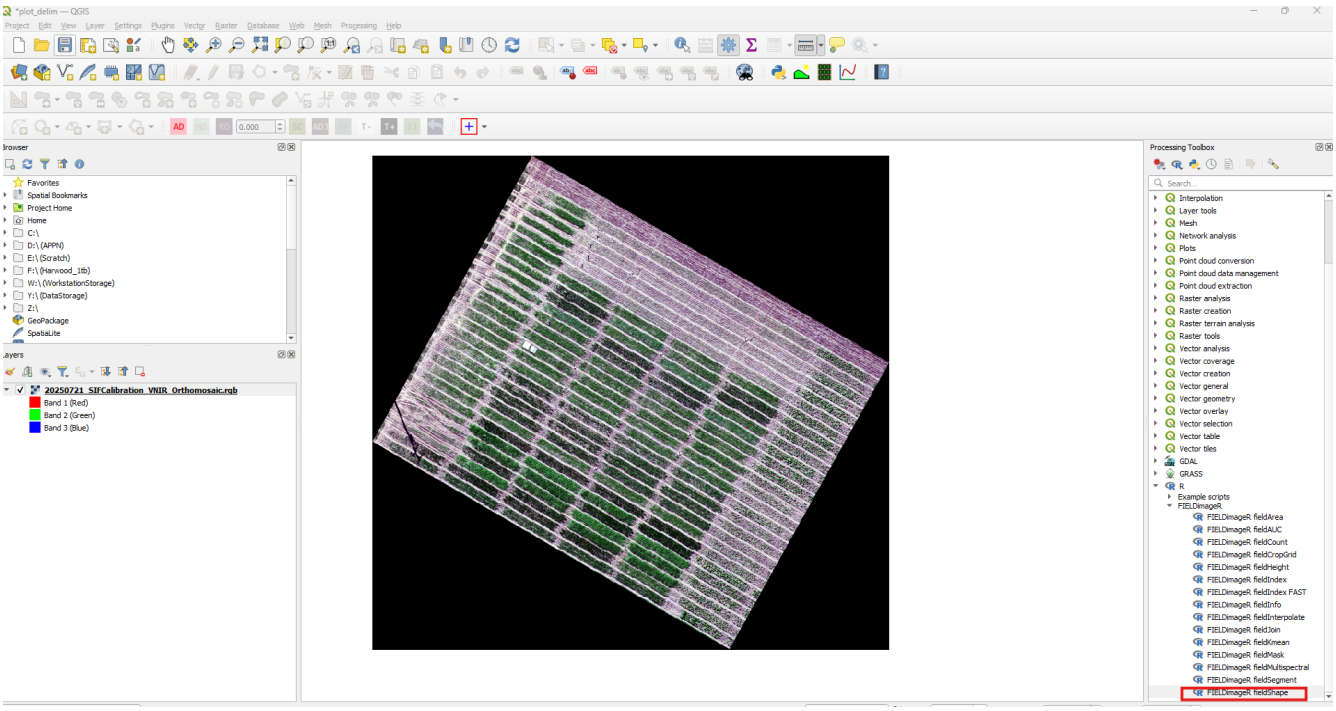
For this plot (2025 USYD Sif Calibration) we know that field techs are not taking any measurements from buffers, and we have been told to assume that there is and edge effect. Therefore, we follow these buffer guidelines:

"APPN minimum buffer (working rule): 0.3 m from each plot end and 0.2 m from each plot side (across the drill direction), or the width of one row — whichever is larger."

Some quick measurments shown below that 0.2 meters may still include some of the buffer, so here we will us "the width of one row"



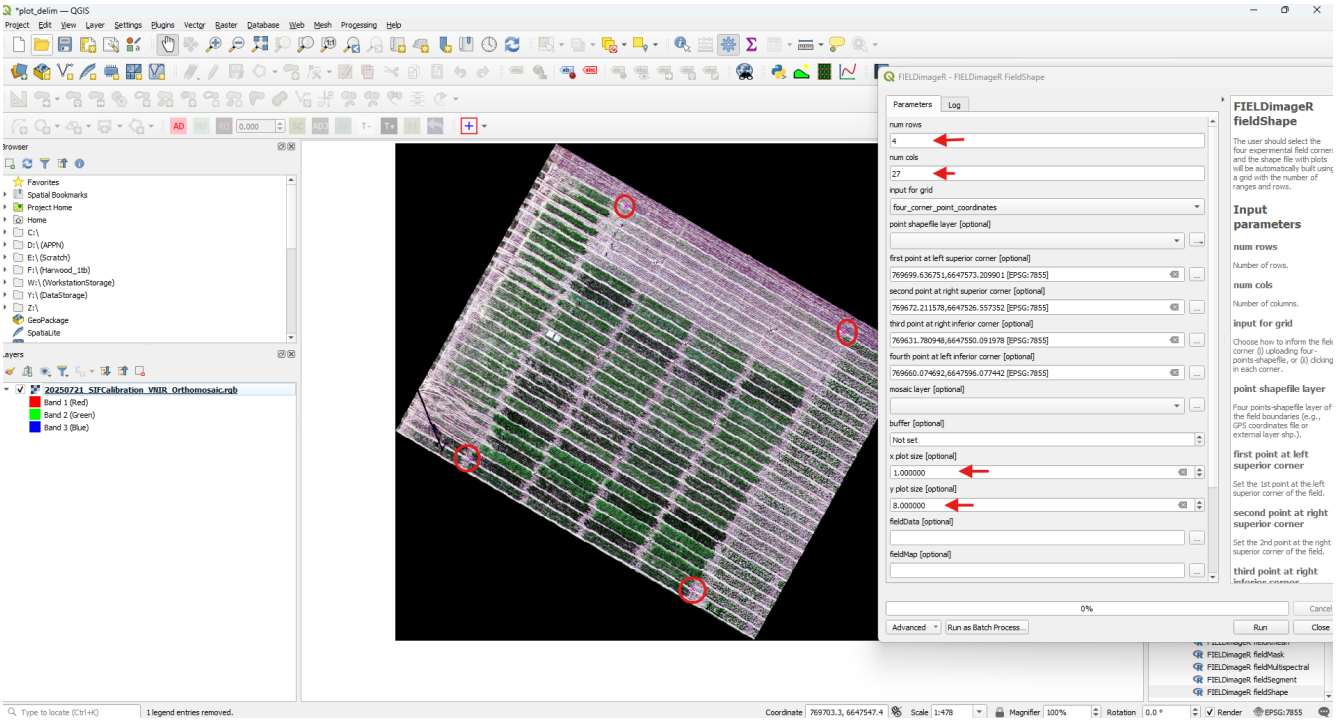
2. Open the **fieldShape** module from the Processing Toolbox.



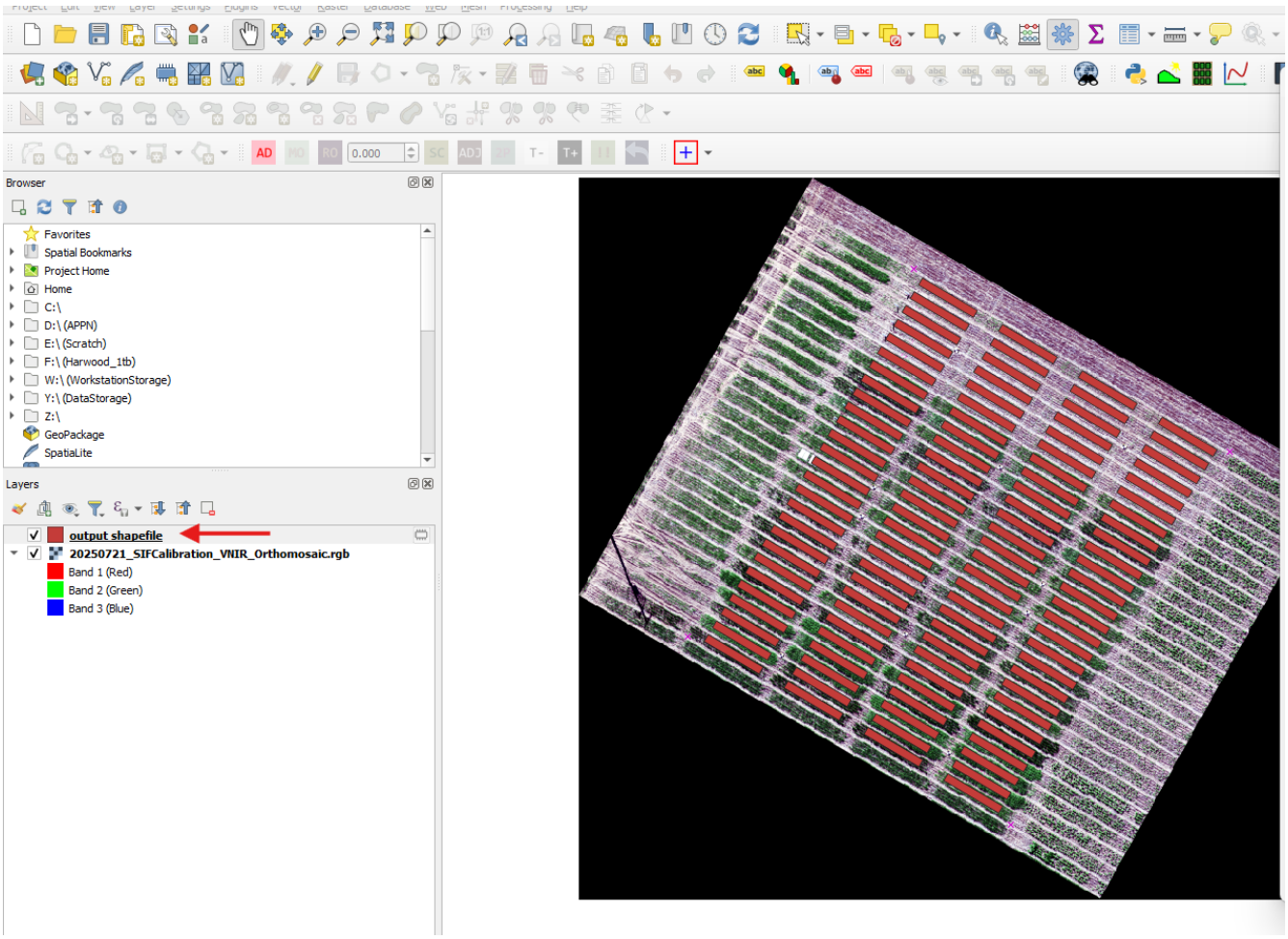
3. Fill in the module parameters:

- Number of **rows** and **columns**.
- The **corners** of the trial area (most critical for an accurate fit).
- **Plot size**.
- **Buffer** — essential for establishing a common analysis area by mitigating edge effects.

> **Note**
>
> You can achieve fine plot delineation by playing with buffer or plot size and the corners. These methods have an element of trial and error in them and involve a lot of manual QC and tweaking.



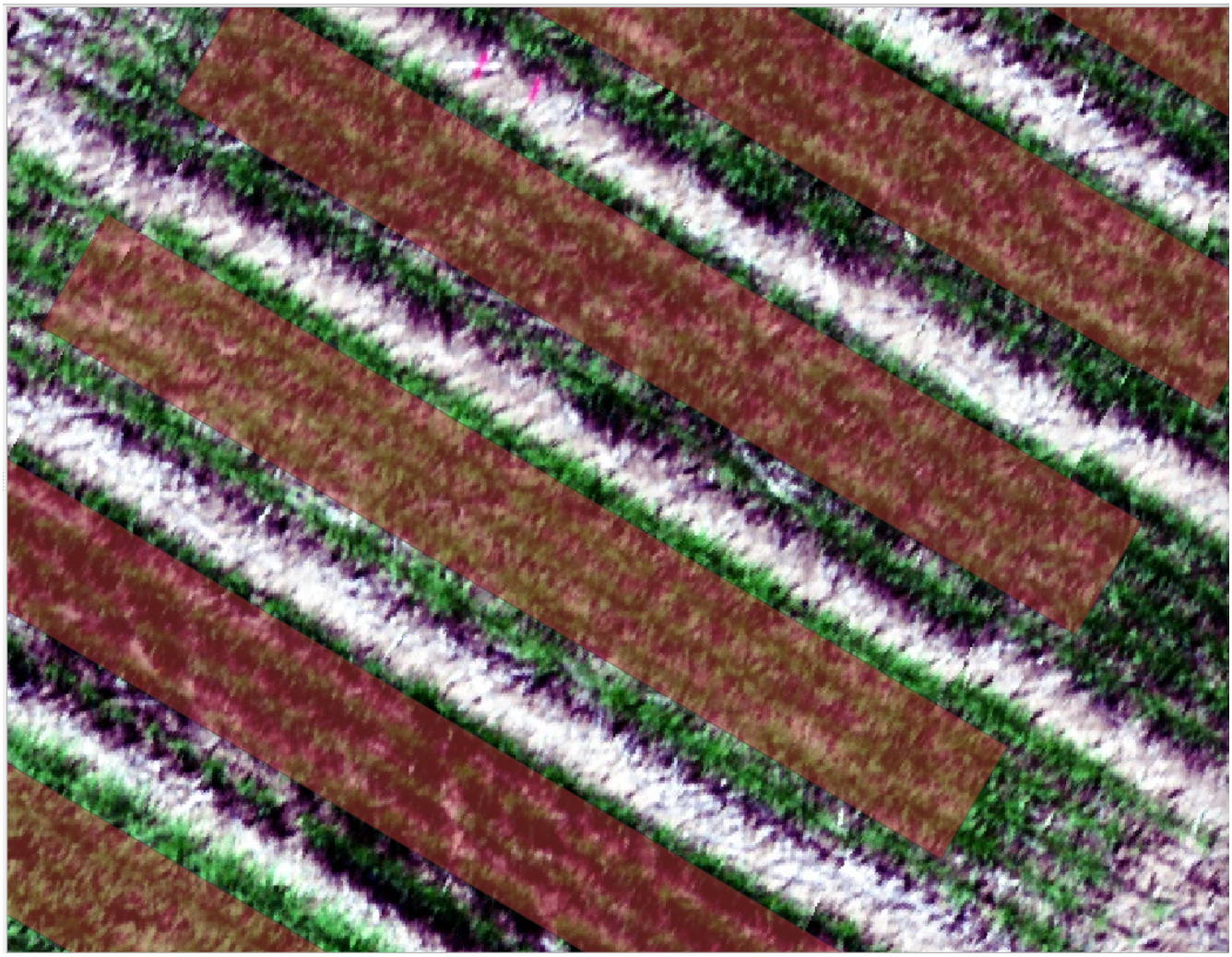
4. Click **Run**.



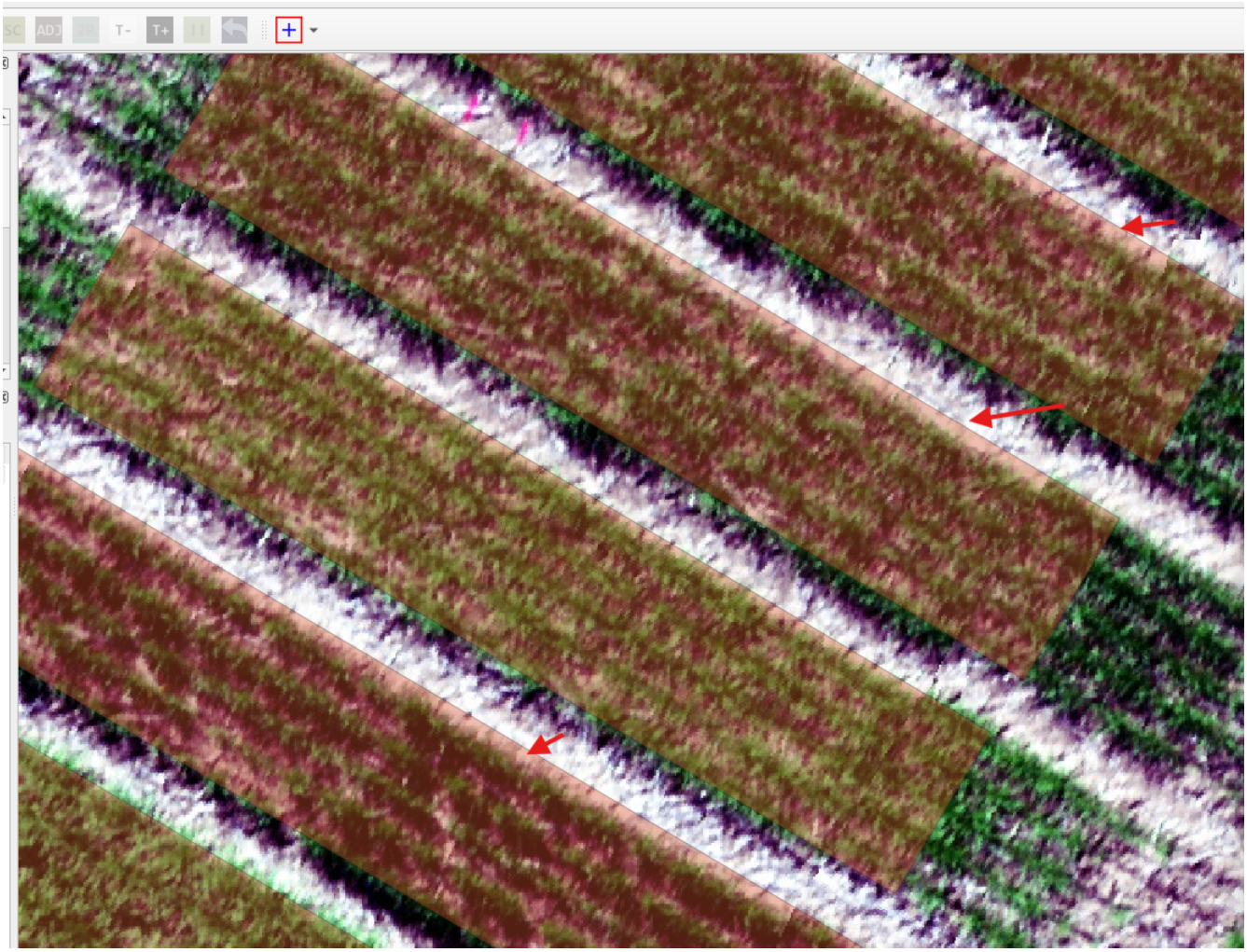
5. The plot shapefile will be generated.

It is now valuable to inspect the some plots and make sure that you are *mostly* not capturing any buffer rows. It's likely some plots will not be perfect, that is fine, however as a rule of thumb you need to a clear chunk of plant material outside your plot segmentation. If in any plots your polygon captures the soil in between plots you must redo your plot delineation workflow

Here is an zoomed in section from the workflow above, where the buffer rows are excluded.



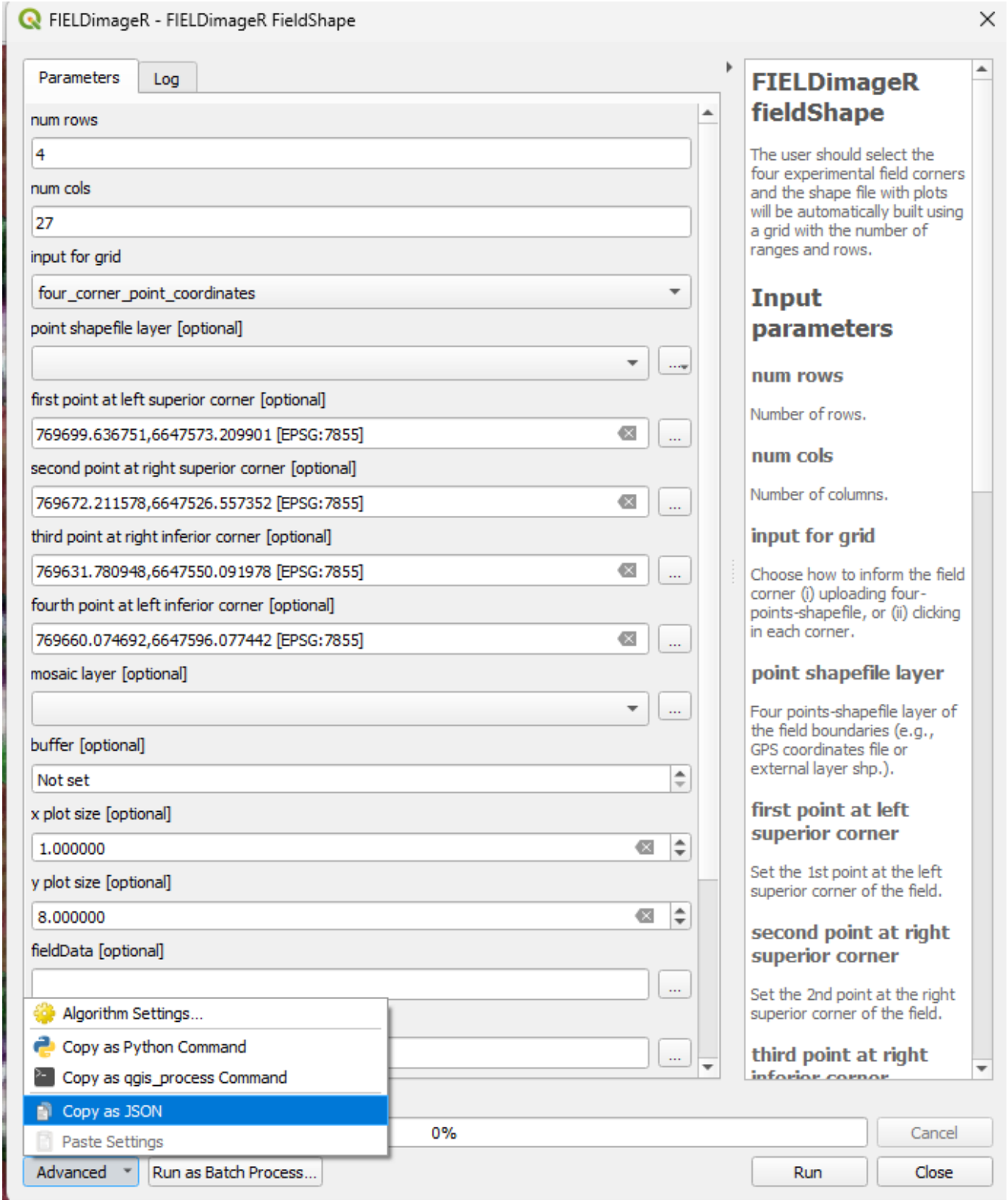
Here is a (bad) example from an early iteration of the workflow where the buffer rows and some soil are clearly in-



cluded.

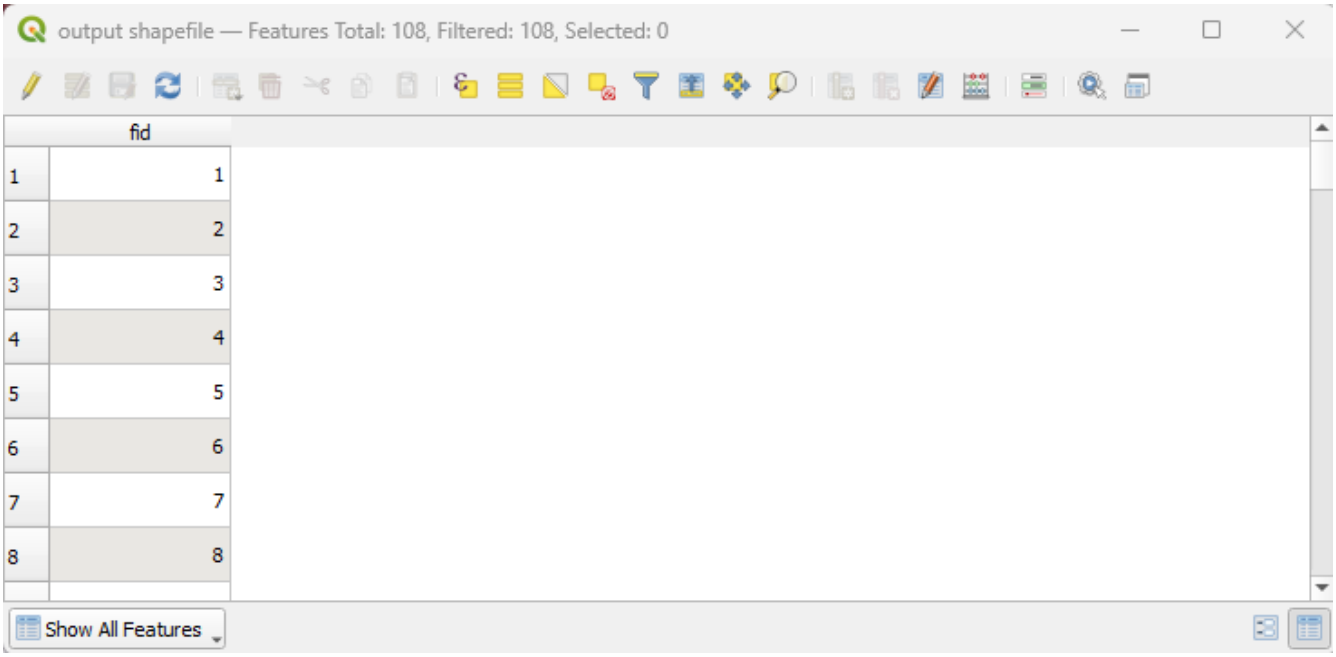
Save your settings Save your fieldShape settings before closing the tool — inputs will be wiped if FIELDimageR is closed.

Save your settings Save your fieldShape settings before closing the tool — inputs will be wiped if FIELDimageR is closed. Use **Copy as JSON** and save the contents as a text file alongside the shapefile in the trial's `Plot_Layout` directory.



Output

The shapefile produced by FIELDimageR contains plots identified only by `fid`.



Attach trial metadata as described in Joining Trial Information, then save the final shapefile to the trial's `Plot_Layout` directory per the APPN Plot Shapefile Standard.

Method 2: DPIRD Field Mapping Tool

The DPIRD Field Mapping Tool is a desktop application for digitising and managing agricultural field trial plot boundaries over drone orthomosaic imagery. Built with Streamlit and Python geospatial libraries, it runs locally in your web browser with no cloud dependency.

Developed at the Department of Primary Industries and Regional Development (DPIRD), Western Australia, as part of the Australian Plant Phenomics Network (APPN).

Features

- **Generate Grid** — Create regular plot grids over drone orthomosaics by drawing a trial boundary and specifying banks, rows, buffer, and plot dimensions.
- **Edit Grid** — Interactive browser-based polygon editor with drag, vertex editing, multi-select, copy/paste, measurements, undo, and keyboard shortcuts.
- **Convert File** — Convert between Shapefile, GeoJSON, GeoPackage, and KML formats with optional CRS reprojection (GDA2020, GDA94, WGS 84, or custom EPSG).
- **Cropping Tool** — Crop rasters (orthophotos, DSMs, GeoTIFFs) to individual plot polygon boundaries, producing one file per plot.

All data is processed locally. No internet connection is required after installation.

Detailed installation instruction and documentation is provided in the DPIRD Field Mapping Tool Documentation within the GitHub repository. Basic guidelines are provided below.

Software Installation

Go to the GitHub repository appndpird/DPIRDFieldMappingTool and download the repository (click **Code** → **Download ZIP**, or `git clone`). The repository contains two platform-specific distributions — choose the one that matches your operating system.

Requirements

	Windows	Linux / macOS
OS	Windows 10+ (64-bit)	Ubuntu 20.04+, Fedora, macOS 11+
Python	Bundled via Miniconda	User-installed Anaconda/Miniconda
Disk space	~2 GB	~2 GB
Internet	First-time setup only	First-time setup only

Windows

1. Extract `DPIRD_Field_Mapping_Tool_windows_v1.6.0`.
2. Place `Miniconda3-latest-Windows-x86_64.exe` in the folder.
3. Double-click `Install_DPIRD_Tool.bat` and press `Y`.
4. Double-click `Run_DPIRD_Tool.bat` to launch.

Linux / macOS

1. Extract `DPIRD_Field_Mapping_Tool_linux_v1.6.0`.
2. Ensure Anaconda or Miniconda is installed.
3. Run:

```
chmod +x install_dpird_tool.sh run_dpird_tool.sh
./install_dpird_tool.sh
./run_dpird_tool.sh
```

The tool opens in your browser at http://localhost:8501.

Generating the Plot Shapefile

1. Open the tool and navigate to the **Generate Grid** tab.
2. Enter your project folder path (the folder containing the orthomosaic `.tif` file). Click **Browse** or paste the path directly.
3. Select the orthomosaic file from the dropdown. Optionally select a CSV file to attach additional plot metadata.
4. Set the grid parameters:

Parameter	Description
Banks	Number of banks (columns) in the X direction.
Rows	Number of rows in the Y direction within each bank.
Buffer (m)	Gap between adjacent plots in metres.
Plot Size (W,H)	Width and height of each plot in metres, comma-separated (e.g. `4,1`).

5. Draw a boundary polygon on the map by clicking the four corners of the trial area in this order:

Top-left (B1R1) → Top-right (BXR1) → Bottom-right (BXRY) → Bottom-left (B1RY)

The first point defines the origin of the grid (plot B1R1). The orientation of the boundary polygon determines the rotation of the generated grid.

6. Click **Generate Grid**. Review the generated grid on the map and in the data table.
7. Fine-tune if needed — changing any grid parameter after generation automatically regenerates the grid using the same boundary.
8. Click **Save Initial Grid** to save as a shapefile.

Plot ID Convention Plots are assigned IDs automatically:

Field	Formula	Examples
`Plot_ID`	Bank × 1000 + Row	B1R1 = 1001, B2R3 = 2003
`B/R`	Bank-Row label	B1R1, B2R3, B12R6
`Bank`	Bank number	1, 2, 3, …
`Row`	Row number	1, 2, 3, …

Editing the Plot Shapefile

After generating (or loading an existing grid via the **Edit Grid** tab), click **Edit Grid** to open the interactive browser-based polygon editor in a new tab. The editor provides:

Tool	Shortcut	Description
Navigate	`Esc`	Pan and zoom. Click a polygon to select it.
Drag Plots	`D`	Drag polygons to reposition. Ctrl/Cmd+Click to multi-select.
Edit Vertices	`V`	Drag individual corner vertices to reshape polygons.
Delete	`X`	Click a polygon to remove it.
Draw New	`N`	Click to place vertices; double-click to close.
Measurements	`M`	Toggle edge length labels (metres) on all polygons.
Copy / Paste	`Ctrl+C` / `Ctrl+V`	Duplicate selected polygons.
Undo	`Ctrl+Z`	Revert the last action (up to 50 steps).
Select All	`Ctrl+A`	Select all polygons.

> **Note**
>
> On macOS, use **Cmd** instead of **Ctrl** for all keyboard shortcuts.

Click **Save Shapefile** in the editor to write the edited grid to disk, or **Export GeoJSON** to download a GeoJSON file.

Converting Vector Data File Formats

Use the **Convert File** tab to convert the output shapefile to GeoJSON or reproject to a different CRS:

1. Load the shapefile via **Browse Input File**.
2. Set the output format to **GeoJSON**.
3. Select the target CRS (typically the CRS of the source orthomosaic, e.g. GDA2020 / MGA Zone 50).
4. Click **Convert & Save**.

Cropping Rasters to Plots

Use the **Cropping Tool** tab to crop the orthomosaic (or DSM) to individual plot boundaries:

1. Load the plot shapefile and the raster file.
2. Choose a save folder and base filename.
3. Click **Crop and Save**.

For a grid with multiple polygons, one raster is saved per plot (e.g. `cropped_1001.tif`, `cropped_2003.tif`). The tool automatically reprojects the vector to match the raster's CRS if they differ.

Output

The shapefile produced by the DPIRD Field Mapping Tool contains plots identified by `Plot_ID` (Bank × 1000 + Row), `B/R`, `Bank`, and `Row`. To produce an APPN-compliant plot shapefile:

1. **Rename** `Plot_ID` to `plot_id` (or add a `plot_id` column mapped from `Plot_ID`) to match the Required attributes specification.
2. **Add** an `fid` column if not already present (sequential polygon identifier).
3. **Convert** to GeoJSON using the Convert File tab if the primary deliverable should be `.geojson`.
4. Attach trial metadata as described in Joining Trial Information, then save the final file to the trial's `Plot_Layout` directory per the APPN Plot Shapefile Standard.

Further Documentation

Detailed installation instruction and documentation: DPIRD Field Mapping Tool Documentation.

Method 3: GPT plot creation tool

The GRYFN Plot Extraction Tool is software that automatically generates plot boundary files for agricultural research trials. This procedure describes the complete workflow from project creation to final GeoJSON export.

Official wiki: https://gryfn.gitbook.io/gryfn-software/gryfn-plot-extraction-tool.

Step 1 — Install the Software

Download and install the GRYFN Plot Extraction Tool from the official website. The licence is required from GRYFN.

Step 2 — Create a New Project

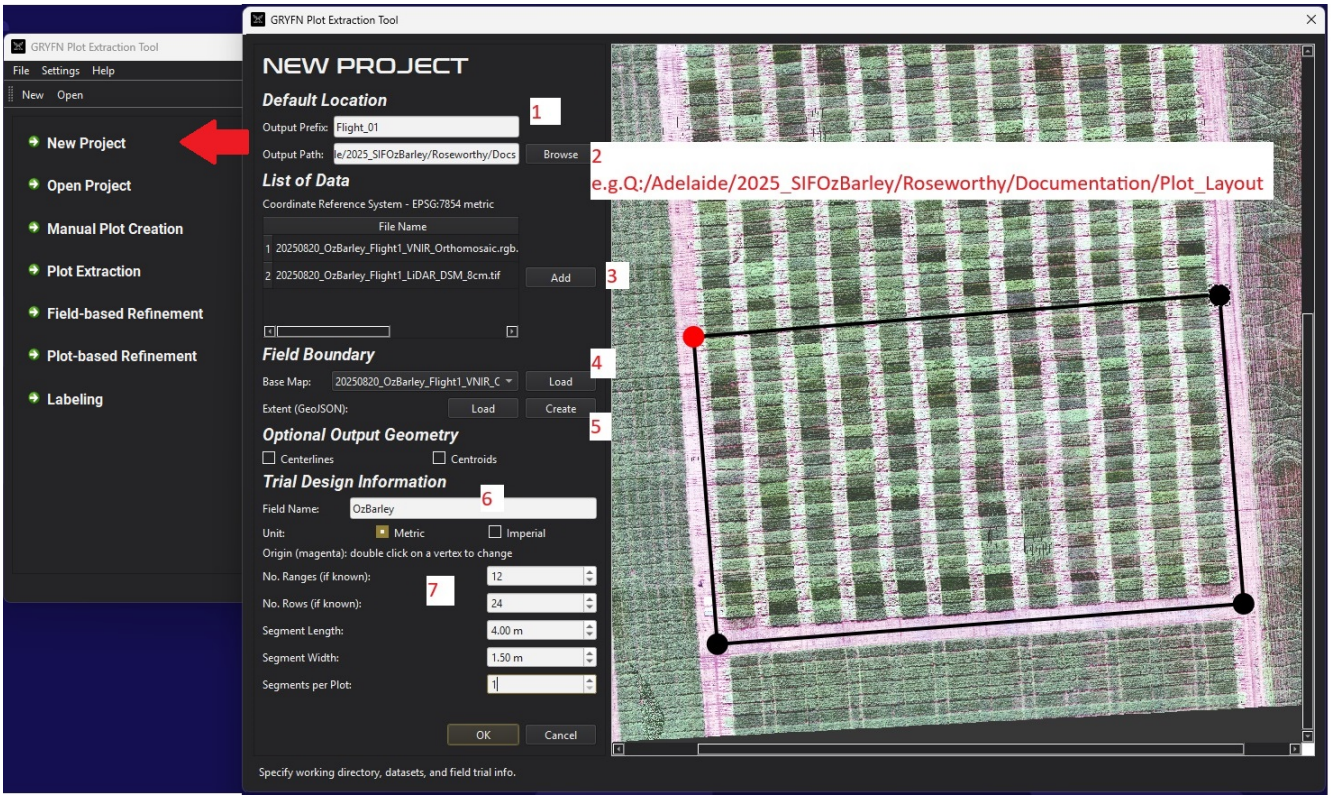
Click **New Project**. A new window will open. Complete the following fields:

- **Output prefix:** Enter a project name.
- **Output path:** Define the folder where outputs will be saved. For example: `Q:\Adelaide\2025_-SIFOzBarley\Roseworthy\Documentation\Plot_Layout`
- **Image layers:** Add one or more image layers. Supported types include LiDAR DSM, high-resolution RGB mosaic, VNIR orthomosaic, and SWIR orthomosaic.
- **Base map:** Select one layer as the base map. The VNIR orthomosaic is recommended due to its smaller file size. Click **Load**.
- **Extent:** Click **Create**. A polygon with four corners will appear on the map. Click and drag each corner to define the site boundary.

> **Note**
>
> Right-click the map to pan, and scroll the mouse wheel to zoom in and out.

- **Field name:** Enter a name for the field.
- **Unit:** Ensure this is set to **Metric**.
- **Plot dimensions:** Enter the range count, row count, plot length, and plot width.
- **Segments per Plot:** If plots contain two or more treatment zones, enter the number of segments. Otherwise, leave it as default: `1`.

Click **OK** to close the window, then click **Open Project** to load the YAML project file you just created.

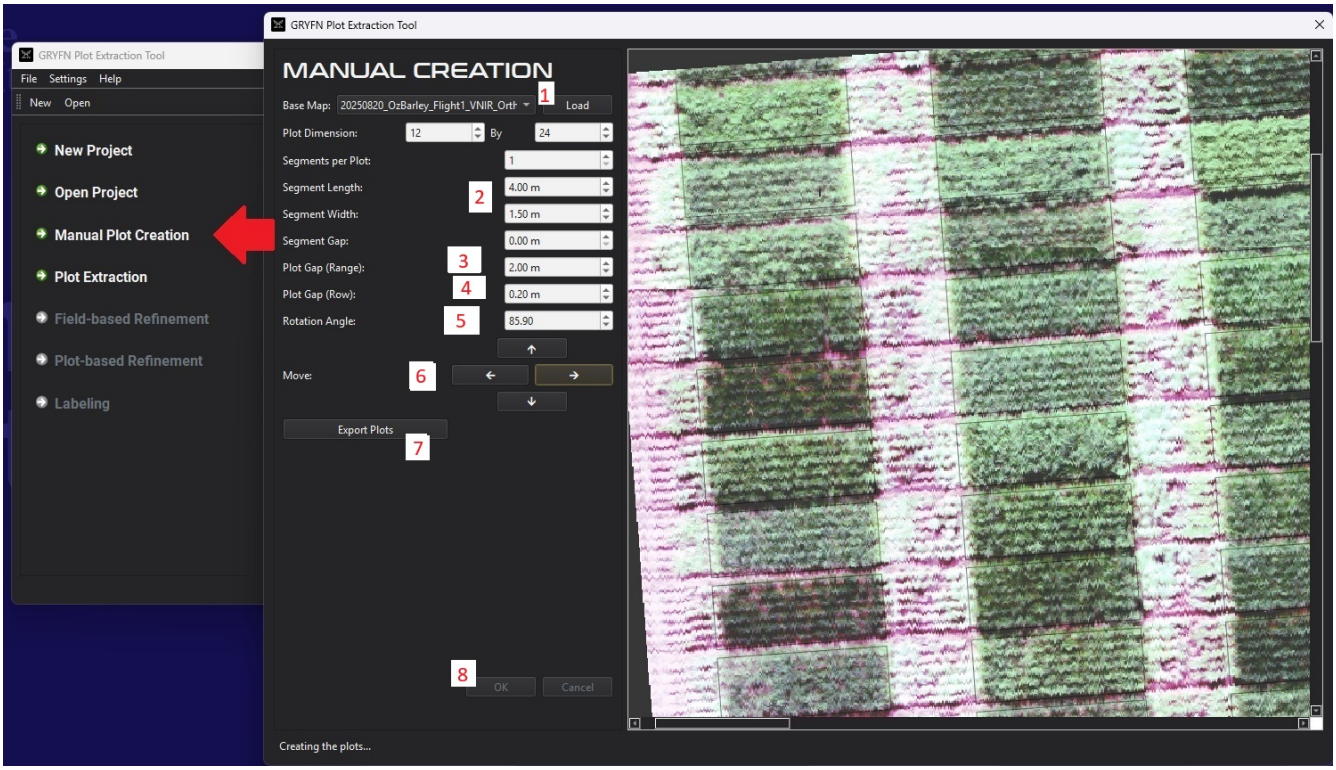


Step 3 — Create the Plots

There are two methods for creating plots: manual and automatic. Both are described below. Choose the method that best suits your data and field conditions.

Option A — Manual Plot Creation This method creates plots with a uniform size and spacing. It is best suited for uniformly planted trials with accurately captured orthomosaic imagery.

- Click **Manual Plot Creation** in the GRYFN Plot Extraction Tool.
- Load the base map (e.g. VNIR orthomosaic) and click **Load**. The map will display with a set of slim plot polygons overlaid, which will need to be adjusted.
- **Plot dimensions:** Enter the plot length and width. Apply standard buffer values (e.g. a 4 m × 1.5 m plot uses a 3.4 m × 1.1 m analysis area).
- **Gap size:** Enter the gap between plots with the appropriate buffer added. Examples:
- Range gap of 2 m: gap = 2 + 0.3 + 0.3 = 2.6 m
- Row gap of 0.2 m: gap = 0.2 + 0.4 + 0.4 = 1.0 m
- **Rotation:** Enter a rotation angle to align the plot grid with the map. Use the up and down arrow keys for fine adjustments.
- **Position:** Use the **Move**, **Up**, **Down**, **Left**, and **Right** controls to align the plot grid with the map.
- Once the plot locations are satisfactory, click **Export Plots**, then click **OK**.

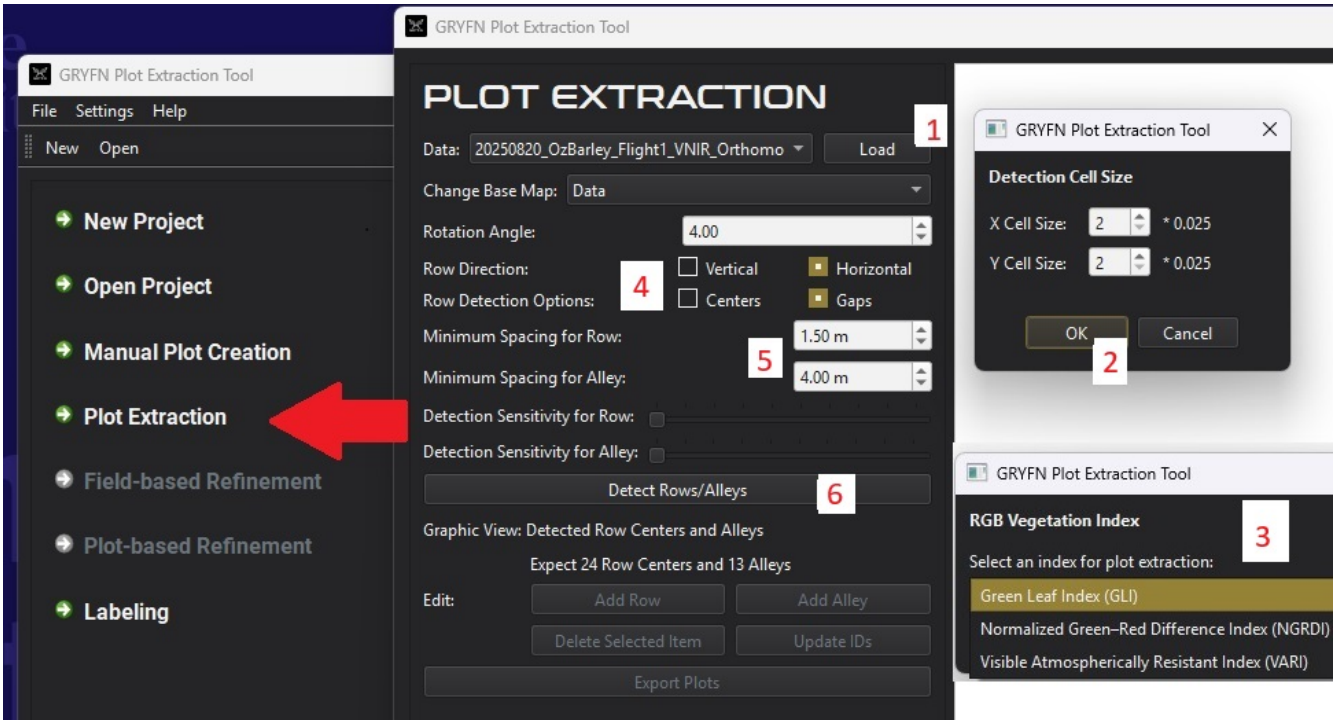


Note

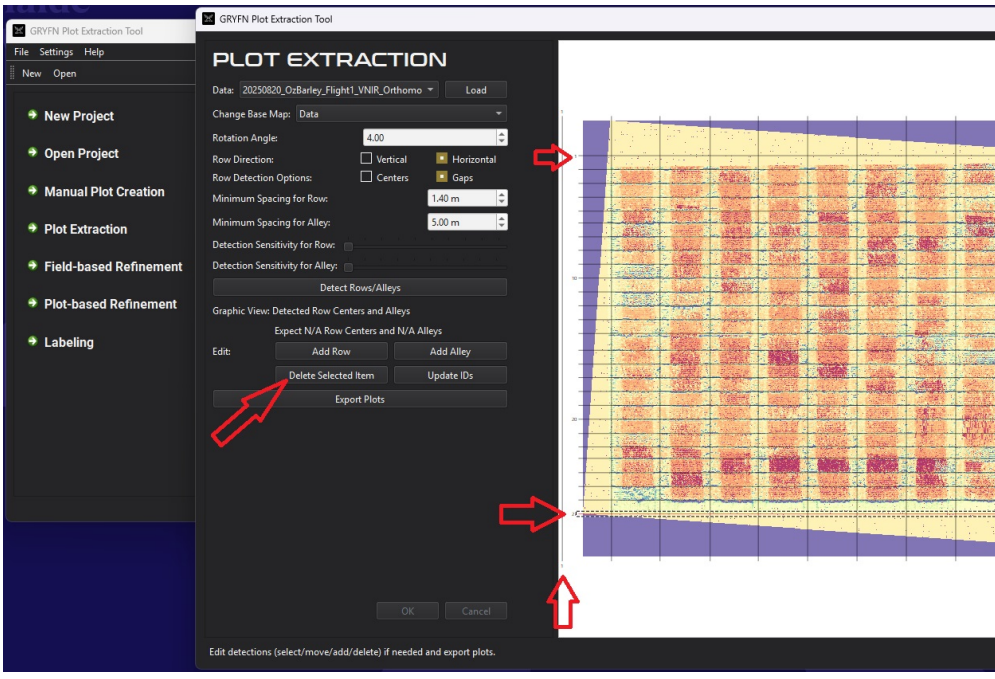
Minor manual adjustment in QGIS or ArcGIS may be required after export. Not all plots are perfectly planted, and GPS drift between flight lines during image collection can introduce small positional offsets.

Option B — Automatic Plot Extraction This method detects plot boundaries automatically using vegetation indices and gap detection.

- Click **Plot Extraction** in the GRYFN Plot Extraction Tool.
- Select the VNIR Orthomosaic as the map and click **Load**.
- Use the default values for detection cell size and click **OK**.
- Select a vegetation index for detection. Both GLI and NGRDI generally produce reliable results.
- A map rotated to vertical or horizontal orientation will appear. Based on the rotation applied, select whether the row direction is vertical or horizontal. Select **Gaps** as the row detection option.
- **Minimum spacing:** Enter the minimum plot dimension (e.g. for a 1.5 m row, enter 1.4 m to avoid missed detections).
- Click **Detect Rows/Alleys**. A grid will appear on the map.



- Verify that the correct number of rows and alleys are present. Extra rows or alleys at the edges can be removed by selecting them and clicking **Delete Selected Item**. Missing lines can be added and dragged to the correct position.

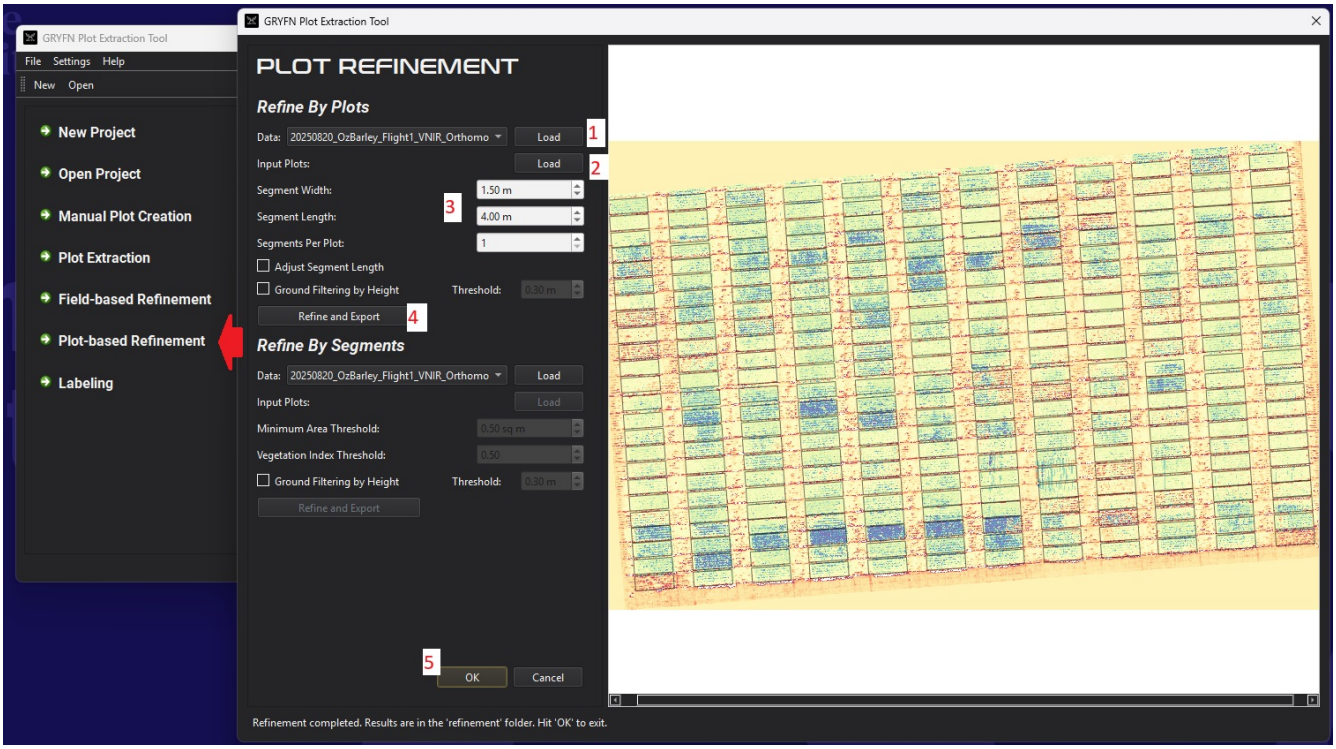


- Click **Export Plots**, then click **OK**.

Plot-Based Refinement (Automatic Method Only) After automatic extraction, an additional refinement step is required:

- Click **Plot-based Refinement** in the GRYFN Plot Extraction Tool.
- Load the map and select a vegetation index.
- Load the input plot file from: `project_name\plot_extraction\project_name_plots_for_refinement.geojson`
- Enter the segment width and length, using standard buffer values.

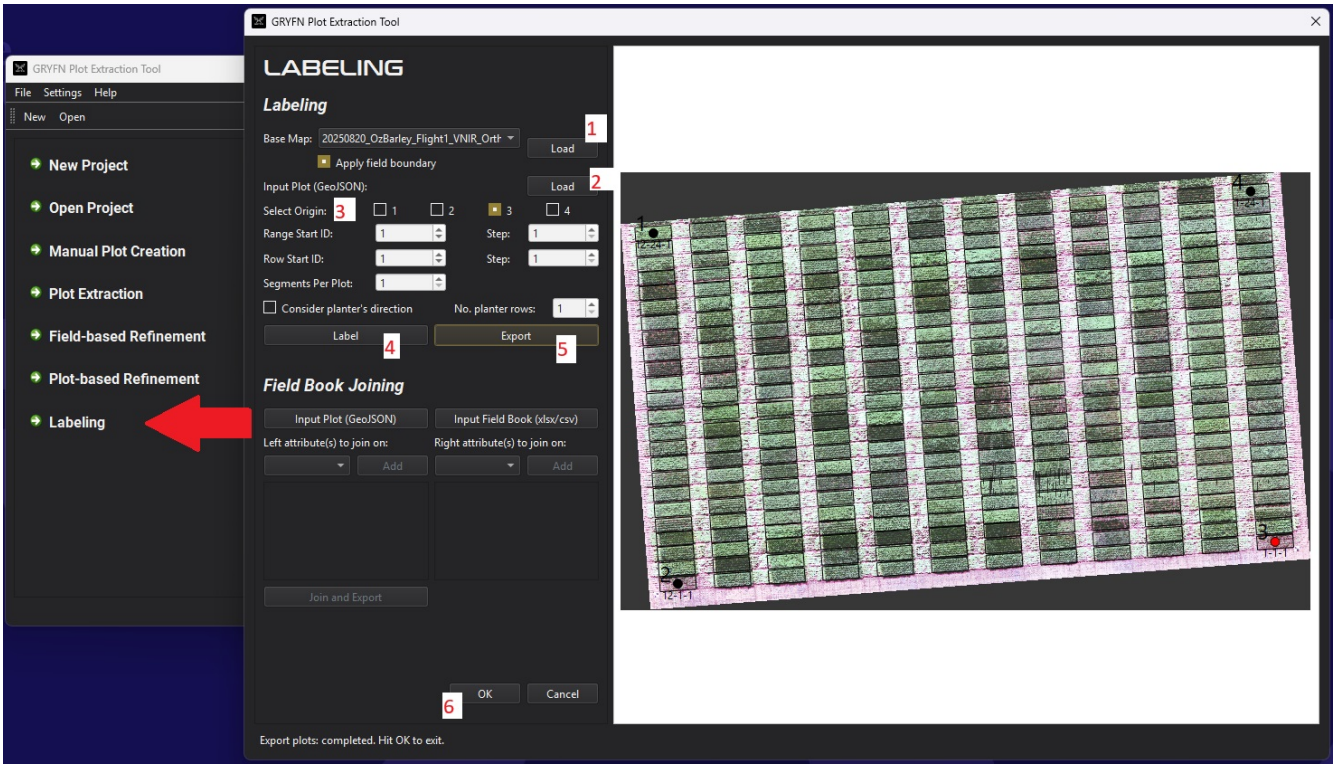
- Click **Refine and Export**.
- Review the output plots to confirm no plots are missing or incorrectly positioned, then click **OK**.



Step 4 — Label the Plots

Click **Labeling** in the GRYFN Plot Extraction Tool and complete the following steps:

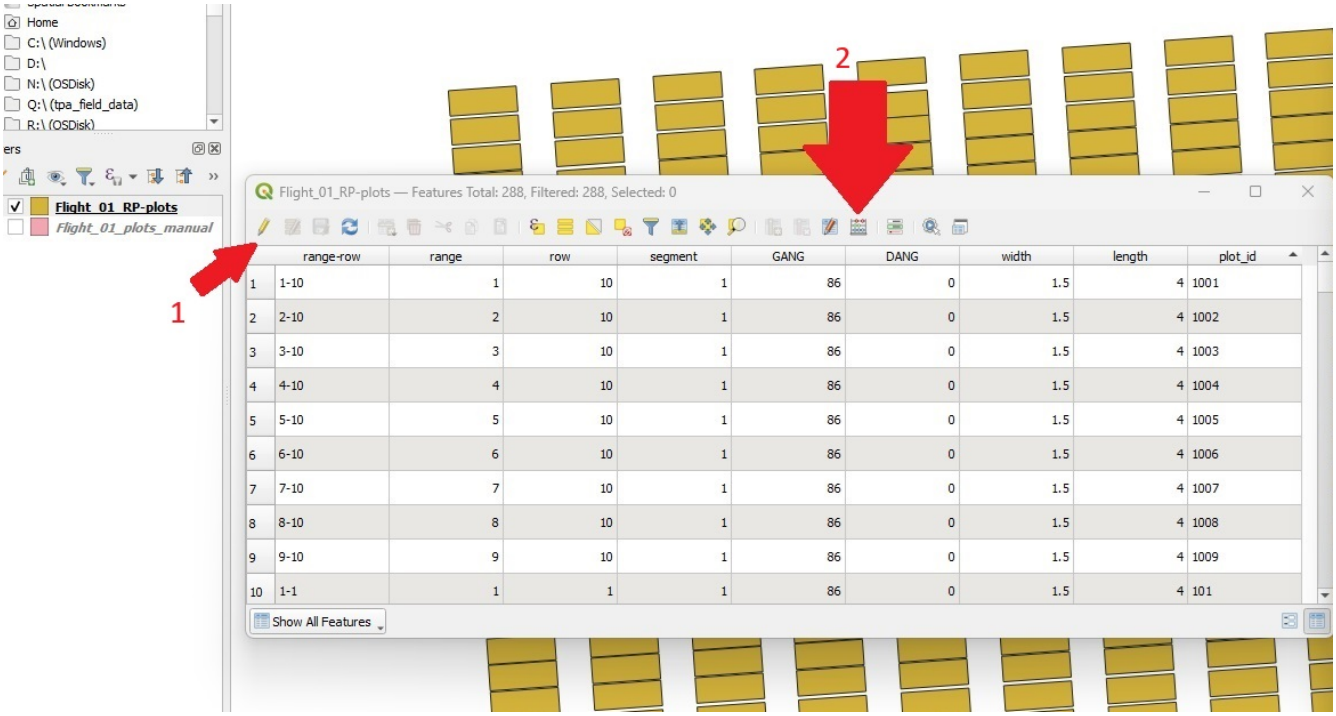
- Load the map and click **Load**.
- Load the GeoJSON file created in the previous step:
- Manual method: `project_name\manual_creation\project_name_plots_manual`
- Automatic method: `project_name\plot_refinement\project_name_RP-plots`
- Select the plot origin (Range 1, Row 1) by choosing position 1, 2, 3, or 4 as shown on the map.
- Click **Label**, then **Export**, then **OK**.



Step 5 — Add `fid` and `plot_id` Columns in QGIS

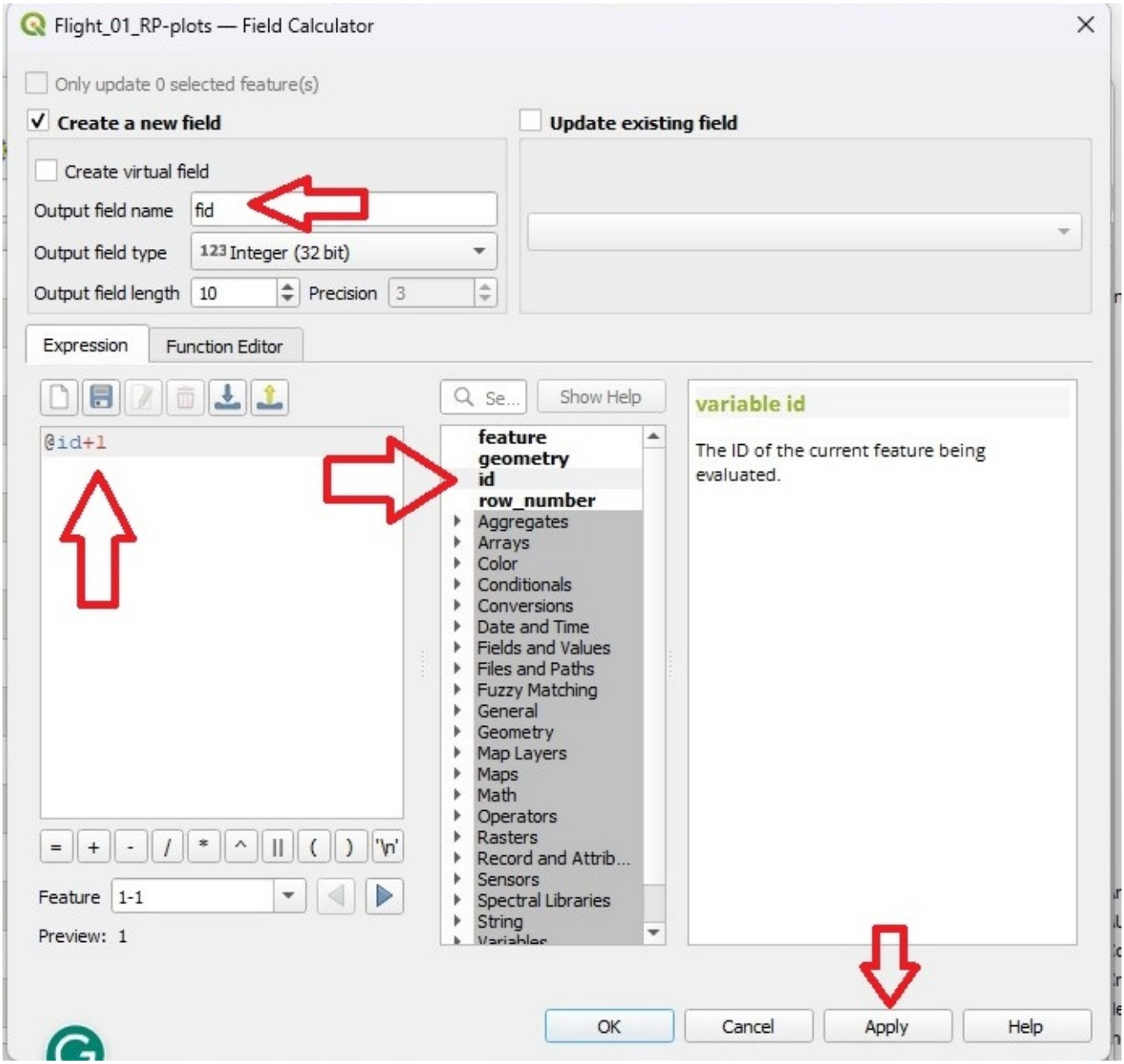
The final step adds two calculated attribute columns to the GeoJSON file using QGIS.

- Open the GeoJSON file in QGIS.
- Right-click the layer in the **Layers** panel and select **Open Attribute Table**.
- Click the pencil icon (top left of the table) to enable editing.
- Click the calculator icon to open the **Field Calculator**.



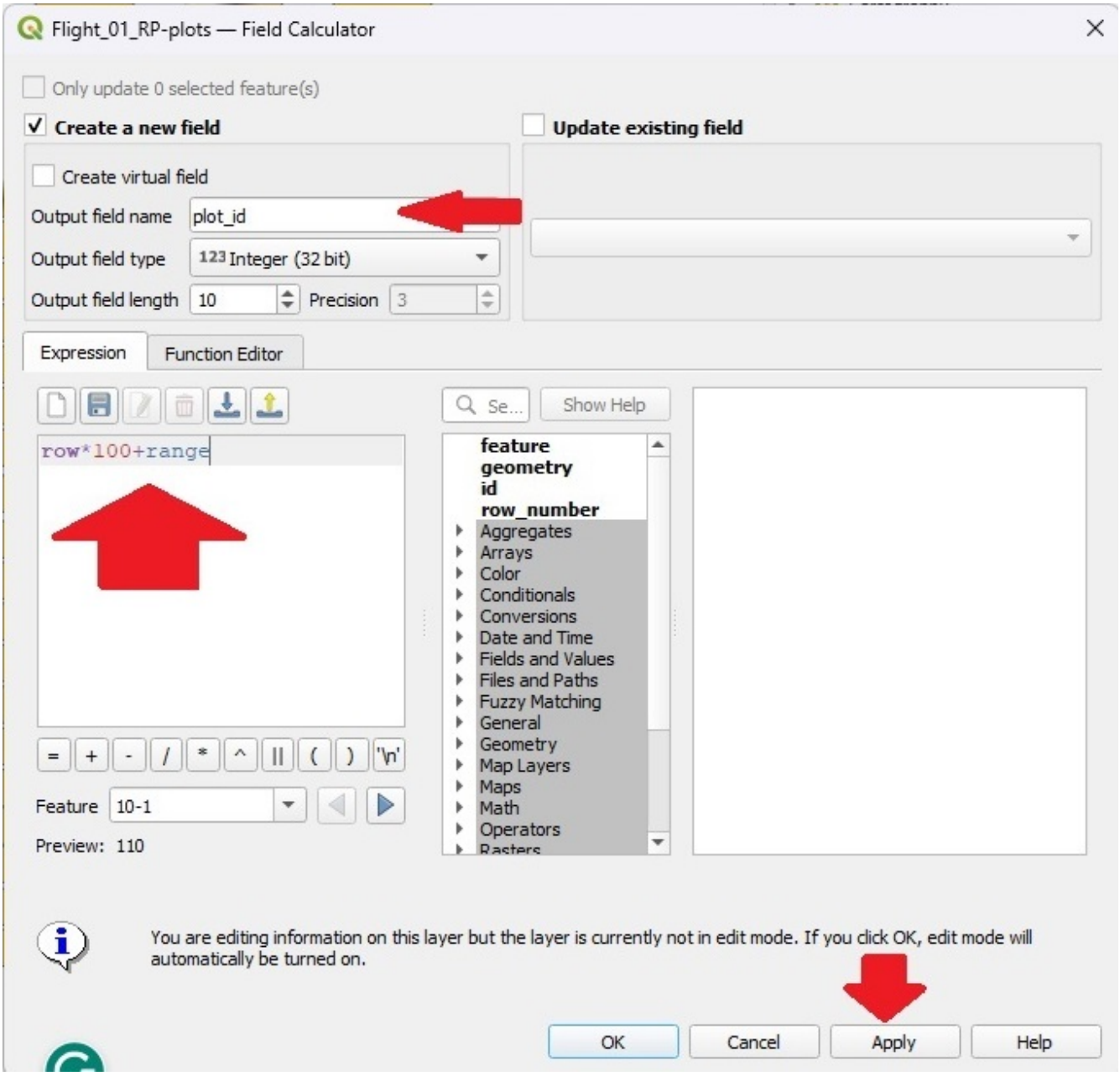
Add the `fid` column:

- **Output field name:** `fid`
- **Expression:** `$id + 1`
- Click **Apply**, then **OK**.



Add the `plot_id` column:

- **Output field name:** `plot_id`
- **Expression:** `row * 100 + range`
- Click **Apply**, then **OK**.
- Click the **Save** button (or press `Ctrl+S`) to save the edits.



Note

Additional plot delineation methods will be documented here as they are adopted by APPN.